Chapter 4
Results and Discussion

Chapter 4 presents empirical results derived from developing and implementing the MedShield Decision Support System (DSS). The system processed sales figures alongside external datasets to perform descriptive, predictive, and prescriptive analyses, supporting a central dashboard linked directly to Specific Objectives 1 through 5 with conditions defined by available records and validation outcomes and outputs impacted by missing records, unmapped entities, or unavailable operational inputs carried explicit limitations (Bello et al., 2026). Contextual constraints substantiated conclusions regarding the role of the developed DSS in advancing inventory planning and minimizing financial losses for MedShield Pharma Corp. across peak and off-peak seasons, accounting for weather dynamics and disease outbreaks within serviced territories (Amuta et al., 2022).

4.1 Presentation of Results

Results produced from preparing, validating, transforming, and analytically processing data used by the MedShield Pharma Corp. Decision Support System (DSS) established the dataset baseline required for subsequent predictive and prescriptive supply chain modeling. Internal sales records and external disease and weather datasets underwent processing through the implemented data pipeline prior to descriptive, predictive, and prescriptive analysis methods (Nguyen et al., 2021). Data quality, coverage, integration capability, and output sample performance determined which factors could support subsequent analytical stages. Items resulting from incomplete or missing records, unmapped entities, unfeasible operational inputs, or unvalidated data were retained with corresponding limitations rather than treated as validated operational outcomes. Subsequent sub-sections outline empirical findings from the multi-stage data pipeline, data quality assessment, feature engineering procedures, and benchmarking modeling efforts.

4.1.1 Data Sources and Coverage

The primary internal data set contained nine years of sales exports supplied by MedShield Pharma Corp., spanning 2017 to 2025. The dataset comprised 59,039 source lines, of which 37,728 transactions entered the analytical fact table after processing and validation, representing 14,076 delivery receipt numbers and 4,561 source product labels. Excluding ten months without reported sales, the nine-year horizon yielded transaction activity across 98 of 108 calendar months. For missing months, the natural collection workflow indicated an absence of records rather than a zero-sales observation.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	MEDSHIELD PHARMA CORP												
2	SALES SUMMARY REPORT												
3	AS OF DECEMBER 2025												
4													
5													
6	Area	DR Number	Date Delivered	Product	Qty	CP	Total CP	Disc	Net CP	TP/UNIT	TOTAL TP	Net Income	%
7	Government	DR RT 0002009	21-Feb-25	PAGBILAO # 411,290	1.00	104,995.00	104,995.00	0.00	104,995.00	213,312.00	213,312.00	-108,317.00	-103%
8	Government	DR RT 0002010	1-Feb-25		1.00	148,236.00	148,236.00	0.00	148,236.00			148,236.00	100%
9	Government	DR RT 0002011	1-Feb-25		1.00	59,740.00	59,740.00	0.00	59,740.00			59,740.00	100%
10	Government	DR RT 0002003-2004	31-Jan-25	PAGBILAO # 2,070,000	1.00	2,069,000.00	2,069,000.00	0.00	2,069,000.00	1,374,000.00	1,374,000.00	695,000.00	34%
11	Government	DR RT 0002005-2006	31-May-25	PAGBILAO # 6,334,470	1.00	6,333,000.00	6,333,000.00	0.00	6,333,000.00	2,916,065.28	2,916,065.28	3,416,934.72	54%
12	Government	DR RT 0002007-2008	1-Feb-25	PAGBILAO # 474,160	1.00	469,000.00	469,000.00	0.00	469,000.00	415,000.00	415,000.00	54,000.00	12%
13	Government	DR RT 0872	21-Feb-25	PAGBILAO # 3,658,000	1.00	764,600.00	764,600.00	0.00	764,600.00	2,618,955.96	2,618,955.96	-1,854,355.96	-243%
14	Government	DR RT 0873	21-Feb-25		1.00	295,544.50	295,544.50	0.00	295,544.50			295,544.50	100%
15	Government	DR RT 0874	21-Feb-25		1.00	2,595,995.00	2,595,995.00	0.00	2,595,995.00			2,595,995.00	100%
16	Admin	DR 025263	3-Jan-25	CIUMADE	7.00	678.57	4,749.99	0.00	4,749.99	450.00	3,150.00	1,599.99	34%
17	Admin	DR 025263	3-Jan-25	SANOMAX - FA	7.00	857.14	5,999.98	0.00	5,999.98	560.00	3,920.00	2,079.98	35%
18	Admin	DR 025263	3-Jan-25	LACTAMOX 500MG/125MG	2.00	2,800.00	5,600.00	0.00	5,600.00	1,115.00	2,230.00	3,370.00	60%
19	Admin	DR 025263	3-Jan-25	MUCOLIFAR 500MG	2.00	3,500.00	7,000.00	0.00	7,000.00	1,100.00	2,200.00	4,800.00	69%
20	Admin	DR 025263	3-Jan-25	ANEROCIN 300MG	2.00	3,800.00	7,600.00	0.00	7,600.00	979.00	1,958.00	5,642.00	74%
21	Admin	DR 025263	3-Jan-25	GLOVES SMALL NITRILE NONSTERILE (SURGITECH)	3.00	450.00	1,350.00	0.00	1,350.00	200.00	600.00	750.00	56%
22	Admin	DR 025264	8-Jan-25	CIUMADE	7.00	678.57	4,749.99	0.00	4,749.99	450.00	3,150.00	1,599.99	34%
23	Admin	DR 025264	8-Jan-25	FERGLOBIN 250MG CAP	7.00	517.86	3,625.02	0.00	3,625.02	325.00	2,275.00	1,350.02	37%
24	Admin	DR 025264	8-Jan-25	APCELAC 500MG HERBAL FOOD SUPPLEMENT	7.00	655.00	4,585.00	0.00	4,585.00	400.00	2,800.00	1,785.00	39%
25	Admin	DR 025264	8-Jan-25	CIXTOR-Z 500mg	7.00	630.00	4,410.00	0.00	4,410.00	210.00	1,470.00	2,940.00	67%
26	Admin	DR 025264	8-Jan-25	TRANEXSAPH 500MG	7.00	1,500.00	10,500.00	0.00	10,500.00	365.00	2,555.00	7,945.00	76%
27	Admin	DR 025264	8-Jan-25	ENCLOXIL 500MG	7.00	1,392.86	9,750.02	0.00	9,750.02	780.00	5,460.00	4,290.02	44%
28	Quezon	DR 025488	2-Jan-25	HYCLENS WOUND SPRAY 2% 120ML BOV	20.00	550.00	11,000.00	1,100.00	9,900.00	392.00	7,840.00	2,060.00	19%
29	Quezon	DR 025489	2-Jan-25	BERCEF 250MG/5ML	20.00	450.00	9,000.00	0.00	9,000.00	180.00	3,600.00	5,400.00	60%
30	Quezon	DR 025490	2-Jan-25	BENERVA PLUS 500MG/50MG/100MG/100MCG	2.00	1,200.00	2,400.00	0.00	2,400.00	615.00	1,230.00	1,170.00	49%

Figure 4.1 Excerpt of raw MedShield Sales Summary Report (December 2025)

Meanwhile, external data origination derived from the Department of Health (DOH) and the Philippine Atmospheric, Geophysical, and Astronomical Services Administration (PAGASA). Structured CSV and Excel datasets acquired via DOH Freedom of Information (FOI) and PAGASA Centralized Analytics and Data Science (CADS) requests served as primary external analytical sources, allowing automated processing and integration with the MedShield data pipeline (World Health Organization, 2024).

Year	Morbidity Week	Onset Date	Region	Province/City	Municipality/City	Cases
2021	1	2021-01-01	REGION IV-A (CALABARZON)	CAVITE	CITY OF CAVITE	1
2021	1	2021-01-01	REGION IV-A (CALABARZON)	CAVITE	CITY OF DASMARIÑAS	3
2021	1	2021-01-01	REGION IV-A (CALABARZON)	CAVITE	CITY OF GENERAL TRIAS	3
2021	1	2021-01-01	REGION IV-A (CALABARZON)	CAVITE	GEN. MARIANO ALVAREZ	1
2021	1	2021-01-01	REGION IV-A (CALABARZON)	CAVITE	NAIC	2
2021	1	2021-01-01	REGION IV-A (CALABARZON)	CAVITE	ROSARIO	2
2021	1	2021-01-01	REGION IV-A (CALABARZON)	CAVITE	SILANG	1
2021	1	2021-01-01	REGION IV-A (CALABARZON)	CAVITE	TANZA	4
2021	1	2021-01-01	REGION IV-A (CALABARZON)	LAGUNA	BAY	1
2021	1	2021-01-01	REGION IV-A (CALABARZON)	LAGUNA	CITY OF CABUYAO	5
2021	1	2021-01-01	REGION IV-A (CALABARZON)	LAGUNA	CITY OF CALAMBA	1
2021	1	2021-01-01	REGION IV-A (CALABARZON)	LAGUNA	CITY OF SAN PABLO	2
2021	1	2021-01-01	REGION IV-A (CALABARZON)	LAGUNA	CITY OF SAN PEDRO	1
2021	1	2021-01-01	REGION IV-A (CALABARZON)	LAGUNA	CITY OF SANTA ROSA	3
2021	1	2021-01-01	REGION IV-A (CALABARZON)	LAGUNA	LOS BAÑOS	1
2021	1	2021-01-01	REGION IV-A (CALABARZON)	LAGUNA	MAJAYJAY	1
2021	1	2021-01-01	REGION IV-A (CALABARZON)	QUEZON	LUCBAN	2
2021	1	2021-01-01	REGION IV-A (CALABARZON)	QUEZON	MAUBAN	1
2021	1	2021-01-01	REGION IV-A (CALABARZON)	QUEZON	SARIAYA	1

Figure 4.2 Excerpt of raw DOH Dengue cases report (CALABARZON snippet, January 1, 2021)

YEAR	MONTH	DAY	RAINFALL	TMAX	TMIN	RH	WIND_SPEED	WIND_DIRECTION
2021	1	1	9.5	25	21.8	92	4	20
2021	1	2	51	26	22.5	94	4	20
2021	1	3	19	26	23	90	3	20
2021	1	4	7.5	29.5	23	89	3	30
2021	1	5	3	29.5	22.5	90	2	20
2021	1	6	13	29	22.5	83	3	40
2021	1	7	5	28.5	22.5	84	4	20
2021	1	8	5.5	27	22.2	89	4	360
2021	1	9	4	25	20.7	90	5	360
2021	1	10	-1	26	20.4	87	5	360
2021	1	11	-1	26	20.5	78	3	20
2021	1	12	-1	26.5	20	82	3	10
2021	1	13	-1	28.2	21.5	83	3	30
2021	1	14	-1	28	22	85	3	10
2021	1	15	1.5	27.5	22	90	3	40
2021	1	16	0	29	23.3	80	3	360
2021	1	17	35.5	27.8	22	86	3	30

Figure 4.3 Excerpt of raw PAGASA Tayabas weather station (January 1-17, 2021)

The original methodology evaluated published PDF reports as potential external data sources, including weekly epidemiological bulletins from DOH and probabilistic forecast reports from PAGASA. However, such documents were not treated as primary inputs to analytical models, which prioritized requested CSV files. PDF records remained preserved as supplementary reference materials for contextual verification and qualitative interpretation.

Table 4.1 Dataset Inventory, Ingestion Sources, and Coverage Horizon

Dataset	Source Provider	Historical Horizon / Coverage	Grain / Entity	Record Count / Status
Internal Sales Records	MedShield ERP / Yearly CSV Exports	2017–2025	Line-item delivery records (CSV / Excel)	59,039 source lines; **37,728 accepted fact records** (candidate dataset)
Epidemiological Surveillance Data	DOH – FOI	2018–2026 (2026 partial)	Disease cases aggregated by month and geographic area (CSV / Excel)	4,484,337 source records; **138,673 cleaned monthly records** (candidate external data)

Meteorological Data	PAGASA – CADS	2017–2024 (2025 unavailable)	Daily station observations aggregated monthly (CSV / Excel)	**57 stations; 161,742 cleaned daily records; 5,314 monthly records** (candidate external data)
Proxy Weather Data	NASA POWER / Open-Meteo	Retrieved according to required geographic coordinates	Monthly gridded weather observations (CSV / Excel)	Proxy dataset; retained separately from official PAGASA observations
Epidemiological Reference Reports	DOH Weekly Epidemiological Reports	Published reporting periods	Weekly epidemiological reports (PDF)	Supplementary reference material; not used as the primary analytical dataset
Seasonal & Probabilistic Rainfall and Climate Forecasts	PAGASA	Published forecast periods	Probabilistic rainfall and climate outlooks (PDF)	Supplementary reference material; not used as the primary analytical dataset

The datasets then served the analytical requirements of the DSS. Sales records provided the main source for historical performance, prioritization, forecasting, and scenario-based planning. Disease and weather records provided candidate external signals for predictive analysis (Martinez, 2018). Inventory positions, expiry records, ordering and holding costs, and actual lead-time records required for operational inventory optimization were not supplied during the study.

4.1.2 Data Profiling and Quality Assessment of Sales Reports

Out of the 59,039 source lines as analyzed in the original sales data, 13,643 lines were reported blank, in addition to a number of lines with report and export headers, report titles and report source summaries. Data files used varied in format according to different reporting years and inconsistencies in the way delivery dates were recorded in some of the reports. Some backfills and restatements in

non-study period and duplication of transactions in other tables occurred in the data (Kimball & Ross, 2013).

Product and area information had 4,561 unique source product labels with discrepancies in product names, strengths, packaging information, and non-Medical purchases, and area values containing both location information as well as customer types (such as Government and Hospital). Complete product master approval meant source product labels remained invalid for validated medicines' stock-keeping units. Coverage of external datasets was partial, with PAGASA datasets available until 2024, as CADS possessed data only up to December 2024, and incomplete 2026 DOH data, since 2026 marked the current year as of the writing of the capstone project. Specified conditions were considered project limitations in analytical methods rather than being arbitrarily imposed (Azevedo & Santos, 2008).

Table 4.2 Data Profiling and Quality Assessment Results

Data Quality Issue	**Result**	**Quality Assessment**	**Analytical Implication**
Blank source lines	13,643	Non-transaction records	Excluded from analytical processing
Invalid transaction identity	4,436	Required review	Not accepted into the analytical fact table
Historical restatements	2,373	Potential duplication/replacement issue	Held to prevent double counting
Historical snapshot duplicates	639	Duplicate records	Excluded

Possible duplicates	87	Required further review	Held from accepted facts
Records before study scope	86	Outside defined period	Held from analytical processing
Future-dated records	8	Invalid temporal observation	Held for review
Missing sales months	10 of 108 months	Incomplete monthly coverage	Treated as unavailable observations rather than zero sales
Product labels	4,561 distinct labels	Inconsistent and unverified product identities	Required further standardization and product-master validation
Area labels	Mixed geographic and buyer classifications	Inconsistent geographic meaning	Limited territory-level integration with external data

According to profiling results, source data exhibited structural quality issues requiring controlled handling prior to analysis. Unchecked factors included invalid data, duplicate records, and out-of-scope entries, which were removed from processing (Kimball & Ross, 2013). Furthermore, months lacking reported data were designated as "unavailable" rather than zero sales. Inconsistent product descriptions and area tags were similarly isolated for systematic validation.

4.1.3 Data Preprocessing and Cleaning Results

Subsequent to data quality evaluation, preprocessing of sales data occurred to prepare the records for analysis. Cleaning included record validation, duplicate management, date standardization,

and disposition techniques as specified in the data preparation phase of CRISP-DM (Chapman et al., 2000). Data satisfying established criteria were kept, while remaining entries were removed or reviewed.

Table 4.3 Data Preprocessing and Cleaning Results

Cleaning Procedure	Records Affected	Action Taken	Result
Blank-line removal	13,643	Excluded non-transaction rows	Removed from analytical processing
Invalid transaction identity handling	4,436	Held for review	Prevented unverified transactions from entering accepted facts
Historical restatement handling	2,373	Held for review	Reduced risk of historical double counting
Historical snapshot duplicate removal	639	Excluded duplicate records	Removed from accepted facts
Possible duplicate review	87	Flagged and held	Preserved for further verification
Out-of-scope date handling	86	Held for review	Prevented records outside the study scope from entering analysis
Future-date validation	8	Held for review	Prevented invalid temporal

			records from entering analysis
Non-transaction row removal	39	Excluded title, header, and summary rows	Removed structural content from transaction data
Historical backfill processing	1,835	Retained using actual delivery dates	Preserved valid historical transactions
Final accepted records	**37,728**	**Retained for analytical processing**	**Accepted analytical fact records**

Following preprocessing, 37,728 records were accepted for analytical processing. Excluded and held records remained fully traceable via assigned disposition status codes rather than undocumented removal. Assessment of monthly sales coverage identified 10 unavailable months during the 2017–2025 observation horizon. Missing observations occurred in May to August and October 2017, January to February 2018, and October to December 2019. Consequently, 98 of 108 calendar months contained recorded sales, representing approximately 90.7% monthly coverage. All 72 months from January 2020 to December 2025 contained active sales observations. Sales records for 2026 were unavailable at the data extraction cutoff and were excluded from historical analysis. Missing months received an unavailable designation rather than zero sales to avoid misrepresenting underlying transaction activity.

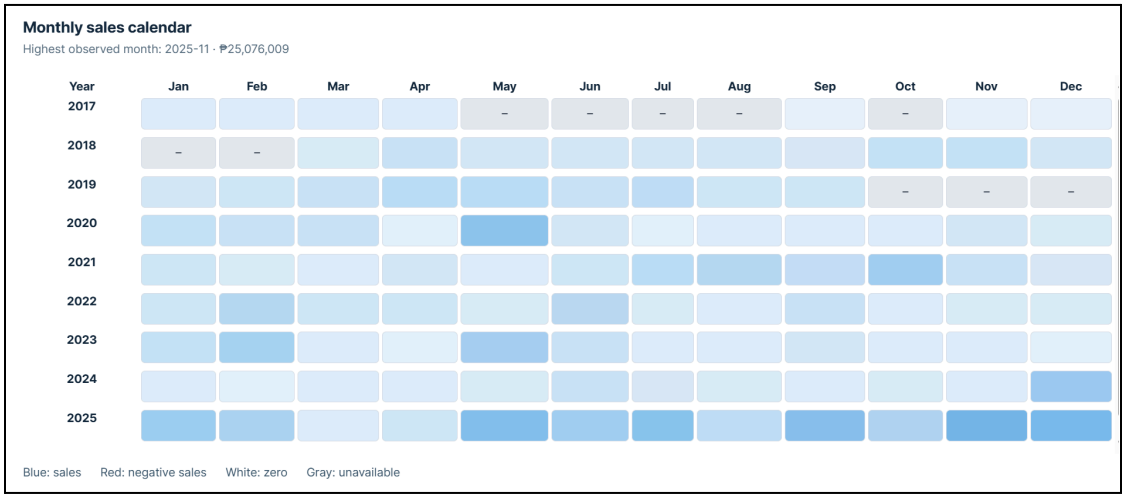


Figure 4.4 Monthly coverage and availability heatmap observing sales (2017-2025)

Figure 4.4 presents monthly sales coverage of MedShield Pharma Corp. from 2017 to 2025. Blue cells represent months with recorded sales, whereas gray cells indicate unavailable observations. The matrix highlights 10 months lacking sales records, concentrated in May-August 2017, October 2017, January-February 2018, and October-December 2019, while continuous monthly coverage occurred from 2020 to 2025. Missing monthly observations remained classified as unavailable rather than replaced with zero values, preserving the distinction between missing data and observed zero sales. Cleaned records established the baseline for transformation, standardization, and analytical integration in subsequent pipeline stages.

4.1.4 Data Transformation and Standardization

Following preprocessing and cleaning, the accepted records were transformed into consistent formats for analytical use. Data transformation focused on standardizing dates, financial measures, product descriptions, customer and area classifications, and other fields required for aggregation and analysis. Feature transformation and consistent data structures help improve the usability of data for subsequent analytical tasks (Dong & Liu, 2018; Kimball & Ross, 2013).

4.1.5 Data Integration and Feature Engineering

Following transformation and standardization, the cleaned datasets were prepared for integration and feature engineering. Internal sales records were aggregated into monthly analytical series, while epidemiological records from DOH-FOI and meteorological records from PAGASA-CADS were independently cleaned and aggregated according to their available temporal and geographic dimensions. NASA POWER weather observations were retained separately as proxy data rather than treated as official PAGASA measurements. Feature engineering supported the preparation of variables required for descriptive and predictive analysis, including monthly periods, sales lags, historical demand measures, and candidate external indicators (Dong & Liu, 2018).

Table 4.5 Data Integration and Feature Engineering Results

Data / Feature	Preparation Performed	Integration Status
Internal sales	Aggregated accepted transactions into monthly analytical series	Ready for descriptive and baseline forecasting
Sales lag features	Generated historical lag variables from monthly sales	Used in predictive evaluation

DOH epidemiological data	Cleaned and aggregated disease observations by time and geographic area	Prepared; territory-level predictive integration pending approved mappings
PAGASA meteorological data	Aggregated station observations into monthly weather measures	Prepared; geographic integration limited by mapping and coverage
NASA POWER data	Prepared gridded weather observations for proxy comparison	Retained as proxy data
Disease-related indicators	Prepared candidate external disease signals	Exploratory; not approved for operational forecasting
Weather-related indicators	Prepared candidate rainfall and weather signals	Exploratory; not approved for operational forecasting
Product and area features	Derived from available standardized classifications	Limited by unverified product and geographic mappings

The integration process produced analytical features suitable for controlled evaluation, though external variables faced specific eligibility constraints. Geographic mappings remained incomplete, PAGASA observations ended in 2024, and the product master lacked final approval. External disease and weather signals were consequently categorized as candidate or exploratory features rather than validated operational predictors.

4.1.6 Analytical Staging and Data Readiness

After transformation and feature preparation, the datasets were staged according to their readiness for descriptive, predictive, and prescriptive analysis. Analytical staging preserved data lineage from the original source records to the cleaned and aggregated outputs. The accepted sales dataset provided the strongest analytical foundation, while external and operational datasets remained subject to additional validation requirements.

Table 4.6 Analytical Data Readiness by DSS Component

Analytical Component	Available Data	Readiness	Remaining Limitation

Descriptive sales analysis	Accepted sales facts and monthly aggregates	**Ready**	Product and area interpretations remain subject to mapping limitations
STL decomposition	Contiguous monthly sales series	**Ready for eligible series**	Series with insufficient continuous observations were excluded
Product prioritization	Standardized source product labels and sales values	**Candidate**	Approved canonical medicine/SKU master unavailable
Territory prioritization	Sales area classifications	**Candidate**	Geographic mappings remain incomplete
Baseline forecasting	Monthly eligible net sales	**Ready for evaluation**	Forecast publication not approved
Disease/weather integration	DOH, PAGASA, and proxy weather data	**Exploratory**	Approved geographic joins and predictive-value gates incomplete
Inventory optimization	Sales history and scenario parameters	**Scenario only**	Actual stock, expiry, lead-time, ordering-cost, and holding-cost inputs unavailable

The readiness assessment indicated that descriptive sales analysis and baseline forecast evaluation could proceed using validated sales records. Product, territory, and external-signal analyses required additional mapping or administrative approval, while inventory optimization remained scenario-based due to missing operational inventory inputs. Such readiness classifications prevented incomplete or exploratory outputs from masquerading as validated operational results.

4.1.7 Model Evaluation and Validation

Predictive models were evaluated against simple forecasting benchmarks before any candidate could be considered for publication. Forecast evaluation followed the principle of comparing candidate methods against appropriate baseline forecasts rather than assuming that a more complex model would provide better predictive performance (Hyndman & Athanasopoulos, 2021). Prophet configurations and

other forecasting approaches were evaluated using historical monthly eligible net sales, while XGBoost-related classification work remained exploratory (Taylor & Letham, 2018; Chen & Guestrin, 2016).

Table 4.7 Model Evaluation and Validation Results

Model / Analytical Method	Evaluation Result	Validation Status
Last-value baseline	Development WAPE of **83.20%**; 2025 audit WAPE of **37.76%**	Best-performing evaluated company-level forecast baseline
Seasonal naive	Evaluated against development periods	Did not outperform the selected baseline
3-month and 6-month means	Evaluated as simple benchmark models	Did not outperform the selected baseline
Additive seasonal ETS	Evaluated as a statistical forecasting candidate	Did not provide sufficient improvement over the baseline
Prophet	Nine sales-only configurations evaluated	Not selected for publication
Disease/weather regressors	Candidate external signals evaluated on available data	No validated improvement supporting operational use
XGBoost classification	Exploratory predictive classification evaluated	Not approved as an operational classifier
EOQ / ROP / safety stock	Calculation logic evaluated using scenario inputs	Scenario validation only
Constrained allocation	Solver output checked for feasibility	Validated as scenario logic, not as an approved procurement recommendation

Forecast evaluation showed that increased model complexity failed to produce sufficient improvement over the simple last-value baseline. The selected baseline recorded a development WAPE

of 83.20% and a 2025 audit WAPE of 37.76%, indicating substantial residual forecast error. Consequently, no advanced forecasting model received approval as an operational forecast. Prescriptive calculations were similarly restricted to scenario validation because actual inventory and procurement inputs were unavailable.

4.1.8 Model Selection and Output Publication

Model selection considered predictive performance, stability, data availability, and readiness for management use. Forecasting methods were retained only when evaluation evidence supported their use, while outputs affected by incomplete mappings or unavailable operational inputs remained classified as exploratory, candidate, or scenario-based. Baseline comparison remained important because forecasting methods should demonstrate sufficient improvement before additional complexity is justified (Hyndman & Athanasopoulos, 2021).

Table 4.8 Analytical Output Selection and Publication Status

Analytical Output	Selection / Status	Basis
Descriptive sales results	**Candidate for publication**	Reconciled against accepted analytical records
STL seasonality results	**Candidate for publication**	Produced only for eligible contiguous monthly series
Historical product prioritization	**Candidate with limitations**	Source product labels remain unverified against an approved product master
Territory prioritization	**Candidate with limitations**	Geographic mappings remain incomplete
Last-value forecast baseline	**Selected evaluation benchmark**	Best-performing evaluated company-level baseline
Advanced forecast models	**Not selected**	No sufficient improvement over the baseline
Disease/weather predictive integration	**Not published**	External integration and predictive-value requirements remain incomplete

Predictive ABC / urgency outputs	**Exploratory**	Insufficient evidence for operational publication
EOQ, ROP, and safety-stock outputs	**Scenario only**	Required operational inventory parameters were unavailable
Constrained allocation	**Scenario only**	Solver logic validated without complete operational constraints

The selection process created a strict distinction between validated analytical evidence and outputs requiring further approval or operational data. Descriptive outputs formed the strongest component of the developed DSS, whereas predictive results remained under evaluation and prescriptive outputs functioned as scenario tools. No model or recommendation lacking adequate validation was designated as an approved operational output. The resulting publication status established the foundation for descriptive, predictive, and prescriptive analyses discussed in Section 4.2.

4.2 Data Analysis

4.3 Dashboard

The MedShield Pharma Corp. Decision Support System was developed to consolidate historical sales information, analytical findings, forecasting evaluations, and scenario-based inventory-planning functions into a centralized interface. The implemented dashboard provided modules for Sales Diagnostics, Product Prioritization, Area Prioritization, Forecast Modeling, and Prescriptive Planning. Supporting interfaces for data management, transaction verification, and weather-data validation were also incorporated to improve transparency and traceability.

The dashboard used analytical records from Databricks Gold candidate views to present historical performance indicators and decision-support outputs. The Executive Overview displayed ₱626,946,742 in observed net sales and approximately ₱288,487,499 in gross profit for the 2017–2025 historical period. Of the 108 calendar months, 98 contained observed sales records, while ten remained unavailable. Missing months were excluded from observed totals rather than replaced with estimated sales values.

The implemented modules were evaluated according to their contribution to the study's analytical objectives and their capacity to support inventory-related decision-making. Descriptive components provided historical sales visibility and preliminary prioritization, while predictive components presented

model evaluation results and limitations. Prescriptive components supported inventory-planning scenarios subject to user-supplied assumptions. Outputs without adequate validation or required operational inputs remained classified as candidate, exploratory, or scenario-based rather than approved management recommendations.

Subsequent subsections discuss implemented dashboard components, observed functions, and corresponding relevance to improving MedShield's inventory planning and reducing potential losses.

4.3.1 Executive Overview

The Executive Overview used historical totals and allowed users to navigate by modules. The module displayed observed net sales of PHP626,946,742 and gross profit of PHP288,487,499 for the period of review spanning 2017 to 2025. The coverage indicator comprised 98 months of observation out of 108 calendar months total. A comparison control needed development through selecting a period prior to reportable matched-period growth to avoid comparing incomplete years.

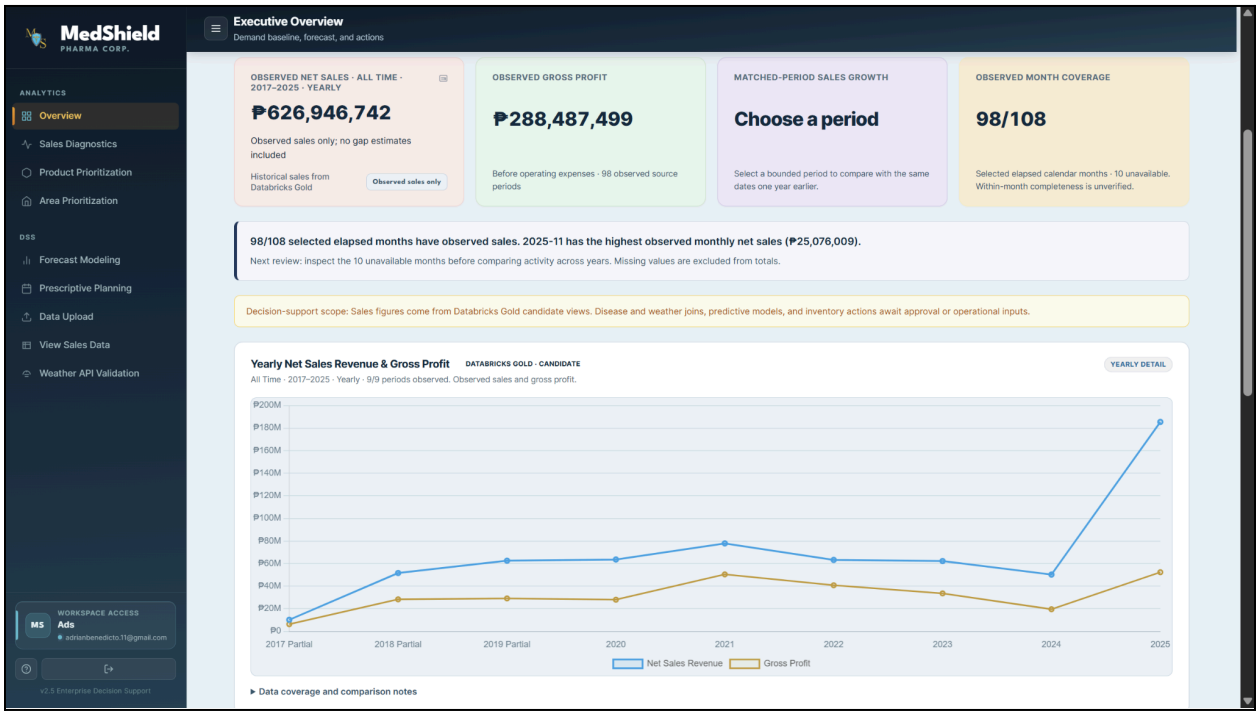


Figure 4.x Executive Overview showing observed sales, gross profit, period coverage, and annual trends

As shown in Figure 4.11, sales peaked at ₱25,076,009 in November 2025. Historical comparisons utilized prior partial years carefully to manage incomplete periods. The 8-year cyclical trend chart demonstrated a sharp volume increase in 2025. Additionally, a status banner explicitly noted that external-signal integrations, predictive models, and inventory measures required formal authorization or missing input parameters.

4.3.2 Sales Diagnostic

The Sales Diagnositic Module provided details on: total annual sales, gross profit, and monthly changes. While the annual representation enabled the comparison of revenues and gross profit, the monthly change graph provided details on how the values were increasing or decreasing. Scope and calculation information was provided for additional clarification of the values.

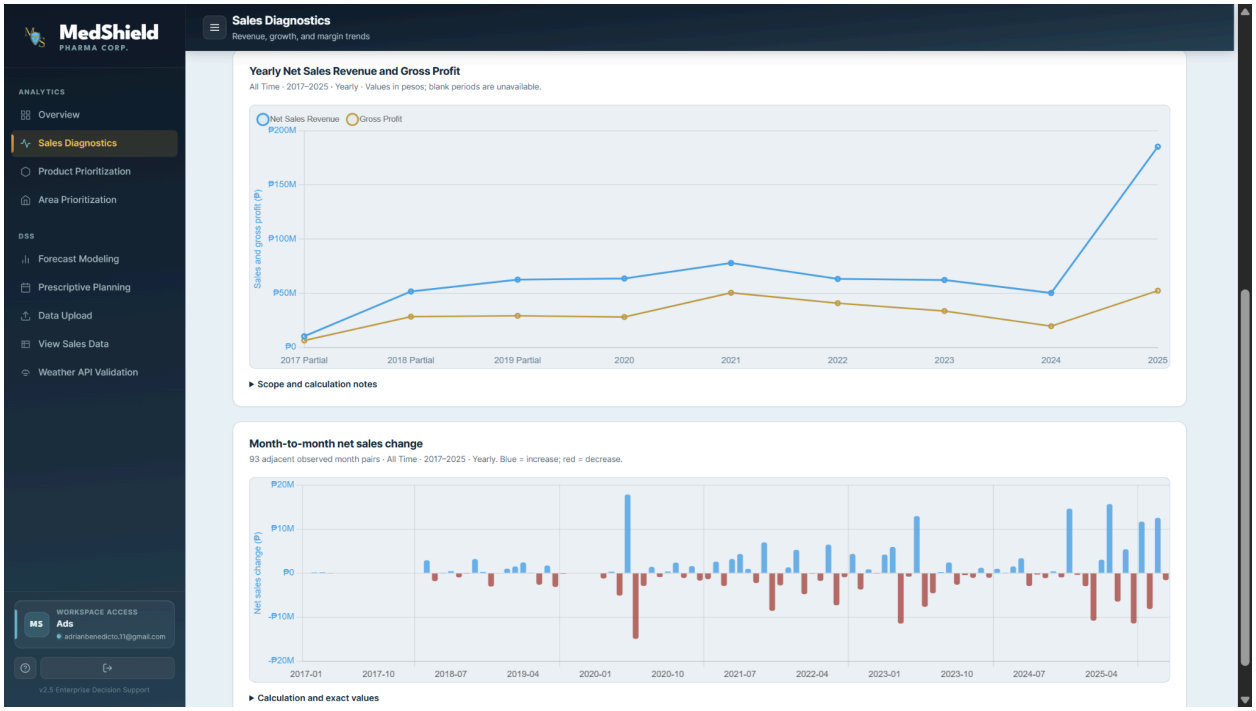


Figure 4.12 Sales Diagnostic Module depicting net revenues per year and monthly net sales changes

Figure 4.12 depicted the Sales Diagnostic Module interface, comprising yearly net revenues alongside gross profit performance and the monthly net sales variation chart. Within the annual representation, the operational association between revenue figures and gross margin outcomes from 2017 through 2025 was highlighted. In contrast, the month-over-month variance chart isolated positive and negative fluctuations through distinct color schemes, highlighting severe shifts in operational performance.

Historical revenue trends demonstrated moderate variations across earlier reporting periods prior to the sharp elevation recorded in 2025. In accordance with empirical observations detailed in Section 4.2.1.1, eligible revenue expanded from roughly ₱50.26 million during 2024 to ₱185.40 million throughout 2025, reflecting approximately 268.9% year-over-year expansion. The indicated surge represented an exceptional deviation from historical baseline trends, emphasizing the necessity for detailed examination regarding underlying transaction breakdowns (Fahad & Kollwitz, 2025).

Potential drivers encompassed expanded institutional procurement, altered fulfillment contracts, shifts in product mix distribution, and variations in historical recording practices; nevertheless, empirical findings remained insufficient to isolate a single verified driver. Consequently, the interface facilitated the detection of unusual sales movements without incorrectly assigning the expansion to specific external drivers or long-term market demand growth.

The monthly variance visualization supplemented annual comparisons by evaluating fluctuations across higher temporal granularities. Upward variations were plotted above the baseline axis, whereas downward movements extended below it. The presentation enabled analysts to pinpoint intervals exhibiting extreme variance and evaluate potential instability within historical demand trends.

Furthermore, the Sales Diagnostic Module incorporated detailed calculation parameters and scope definitions. Supplementary documentation panels advanced process transparency, permitting stakeholders to verify analytical interpretations and underlying financial measurement rules, supporting descriptive research objectives by transforming processed sales facts into actionable margin and revenue insights. Regarding inventory management, findings assisted decision-makers in identifying windows of fluctuating commercial activity that warranted deeper review; however, direct operational metrics such as stockouts, overstock levels, or product expiry losses could not be determined due to missing operational inventory inputs.

4.3.3 Product Prioritization Module

The Product Prioritization Module was developed to examine the contribution of individual source product labels to historical sales performance. The interface incorporated product rankings, candidate ABC classifications, cumulative sales contribution, product activity coverage, and a detailed performance table (Flores & Whybark, 1986; Juran, 1954).

4.3.3.1 Product Contribution and ABC Candidates

Figure 4.13 presents the Product Prioritization Module, including the highest-contributing source product labels and the distribution of candidate ABC classifications. The product contribution chart displayed positive sales and returns, allowing users to distinguish sales generation from negative adjustments associated with returned transactions.

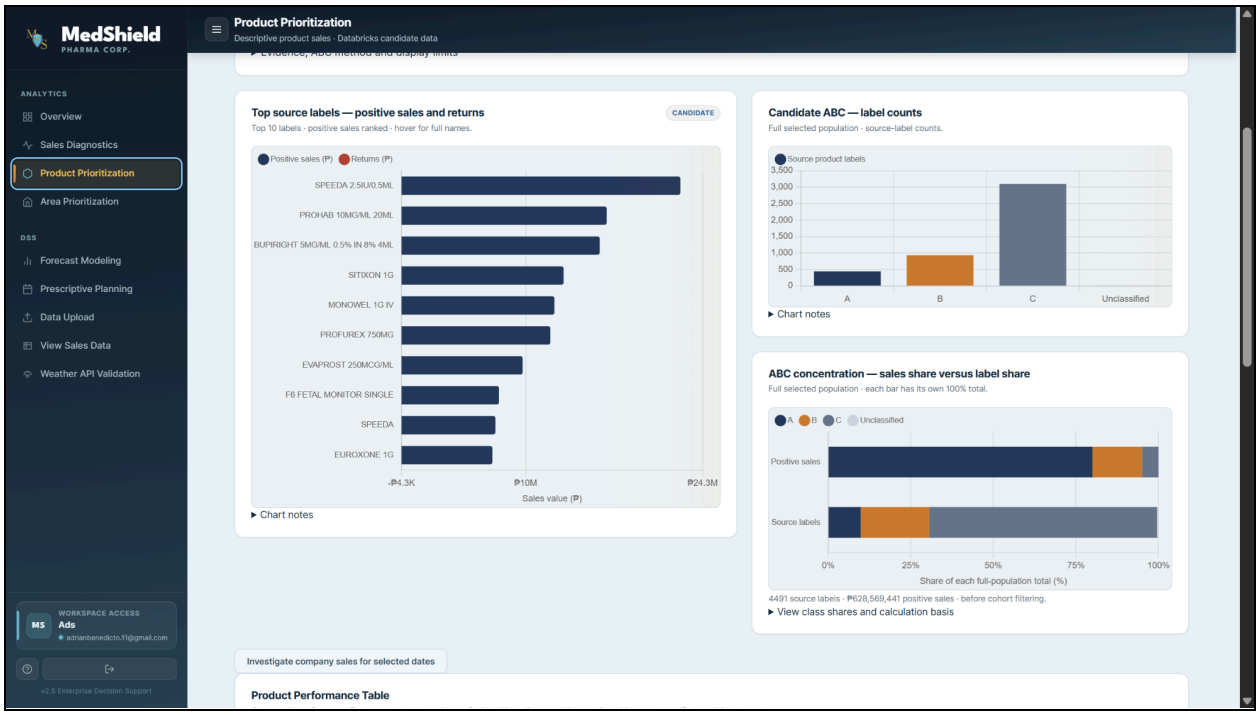


Figure 4.13. Product Prioritization Module Showing Product Contribution and Candidate ABC Classification

The chart identified several high-contribution source labels, including SPEEDA 2.5IU/0.5ML, PROHAB 10MG/ML 20ML, and BUPIRIGHT 5MG/ML 0.5% IN 8% 4ML. The ranking demonstrated how the implemented DSS could identify individual source labels contributing relatively large amounts to recorded sales. The candidate ABC chart presented the number of source labels assigned to the different historical sales-contribution classes. A separate concentration chart compared the share of positive sales for each class against its corresponding share of source labels (Ramanathan, 2006). The results demonstrated how a relatively limited group of high-contribution labels could account for a substantial proportion of historical sales (Juran, 1975).

4.3.3.2 Product Performance Table

The module also contained a detailed Product Performance Table. Figure 4.14 presents the Product Performance Table, which displayed source product labels, net sales, positive sales, returns, contribution percentages, cumulative contribution, candidate ABC classifications, and observed months. The interface also included a cumulative contribution chart identifying the number of source labels required to reach specified percentages of positive sales.

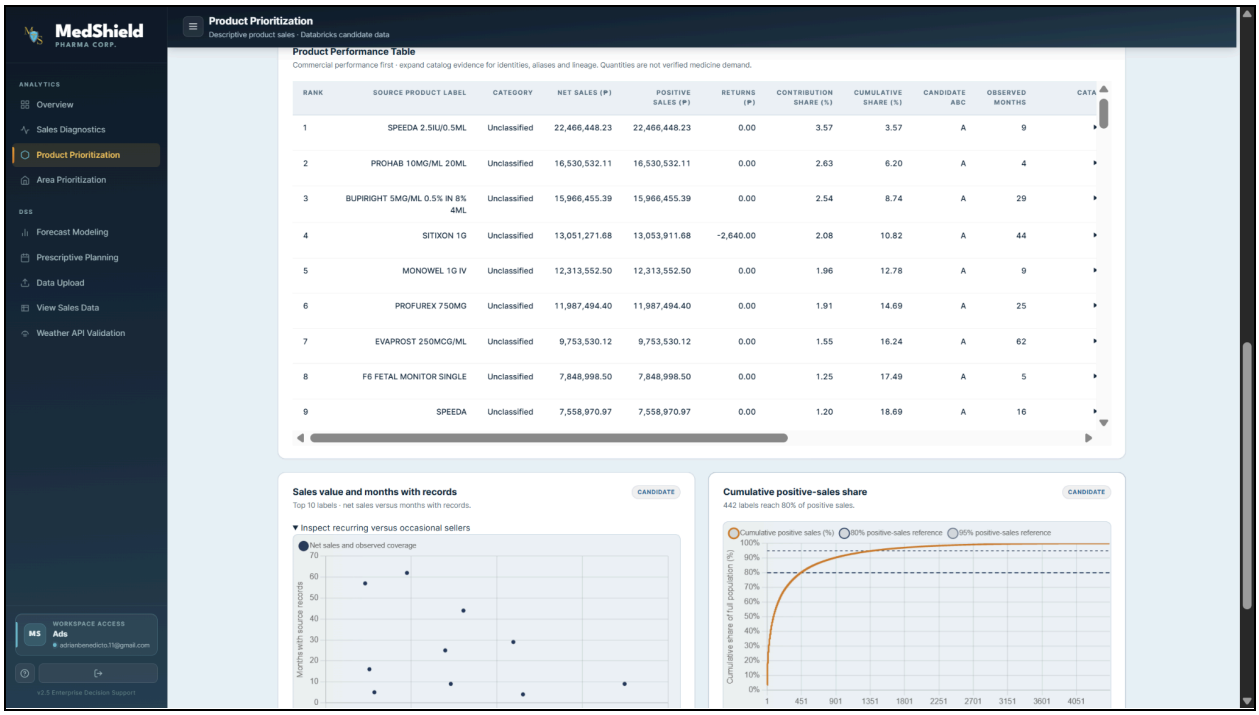


Figure 4.14. Product Performance Table and Cumulative Sales Contribution Analysis

The displayed cumulative contribution analysis identified 442 source labels as accounting for approximately 80% of positive sales within the selected population. The finding indicated an uneven distribution of historical sales contribution across source labels. The observation could support preliminary review of commercially significant products, although revenue contribution alone could not establish inventory urgency or clinical priority. An additional visualization compared sales value with the number of months containing recorded transactions. The comparison helped distinguish labels associated with recurring activity from labels with high sales contributions over relatively few observed months. A product with high sales but limited historical activity could represent a large occasional order rather than consistently recurring demand.

The Product Prioritization Module contributed to the research objective of identifying products for inventory planning by organizing historical sales contributions and transaction activity into accessible indicators. However, generated classifications remained candidate results because source product labels lacked complete validation against an approved medicine-level stock-keeping unit master. Available records contained medical equipment and non-medicine items, further restricting the interpretation of product-level rankings as medicine-specific inventory priorities. Accordingly, the module established a basis for reviewing product contribution and historical activity but did not provide validated medicine demand forecasts or approved purchasing priorities.

4.3.4 Area Prioritization Module

The Area Prioritization Module included mapped and unmapped sales indicators, regional-group summaries, and sales distribution charts which was implemented to examine geographic sales distribution, regional concentration, client-related classifications, and candidate area rankings. The interface provided visualizations and ranking tables intended to support the identification of geographic areas requiring management attention (Amuta et al., 2022; Tiongson & Kummer, 2020). The Area Prioritization Module contributed to the research objective of identifying target locations for inventory distribution. Results provided preliminary geographic concentration metrics while exposing unmapped sales and classification boundaries. Reliable regional allocation recommendations require approved client-to-location mappings, product-level demands, and verified stock availability.

4.3.4.1 Regional Sales Concentration

The module's displayed selection reported approximately ₱89.9 million in geographically mapped net sales, while approximately ₱537 million remained outside the geographic rankings because the source records lacked sufficient geographic attribution under the selected classification rules. The interface therefore separated geographically attributable sales from records without an approved or usable geographic mapping. The separation was important because certain source area values represented buyer categories, such as Government or Hospital, rather than physical locations (Abrigo et al., 2021).

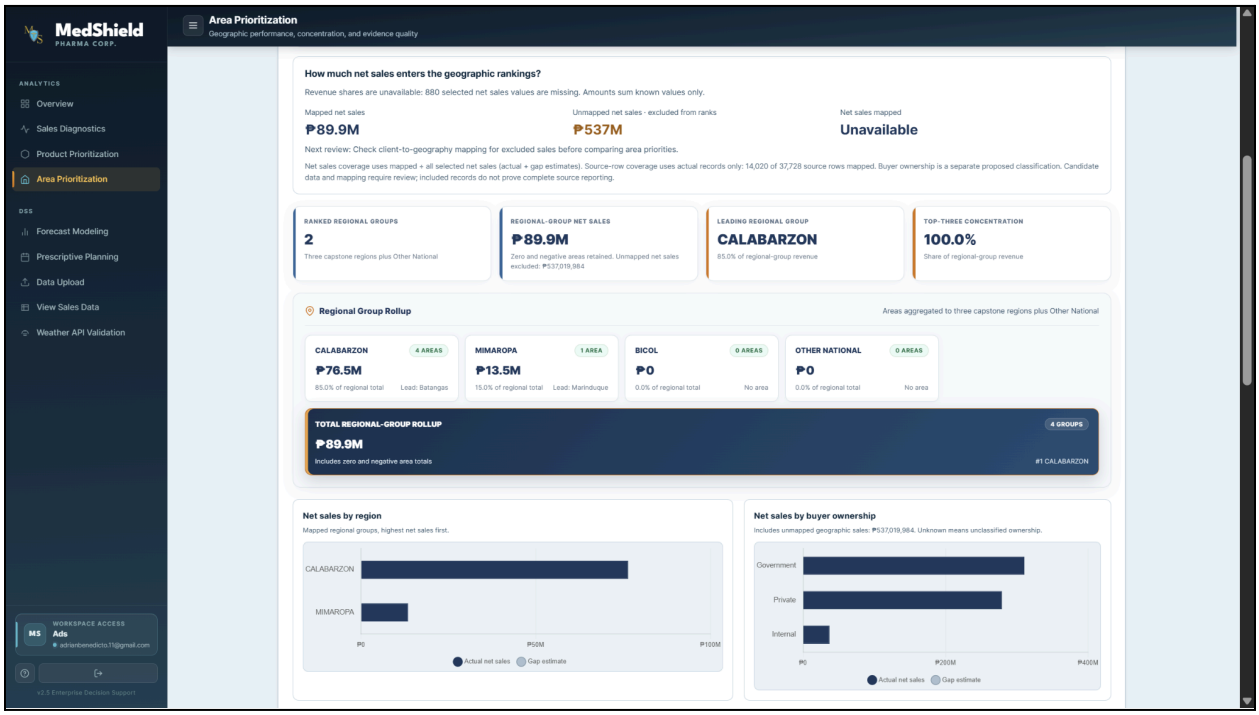


Figure 4.15. Area Prioritization Module Showing Regional Sales Concentration

Within the mapped regional-group results, CALABARZON accounted for approximately ₱76.5

million, representing around 85% of the displayed mapped regional sales. MIMAROPA accounted for approximately ₱13.5 million, or 15%. The displayed regional-group results represented the selected mapped population rather than the complete company sales population.

4.3.4.2 Area Ranking

The interface also contained an area-ranking table with geographic location, regional classification, score, actual sales, estimated values, revenue share, active periods, and evidence status as expressed in Figure 4.16. The table identified areas such as Batangas, Quezon, Marinduque, Laguna, and Cavite among the displayed ranked locations.

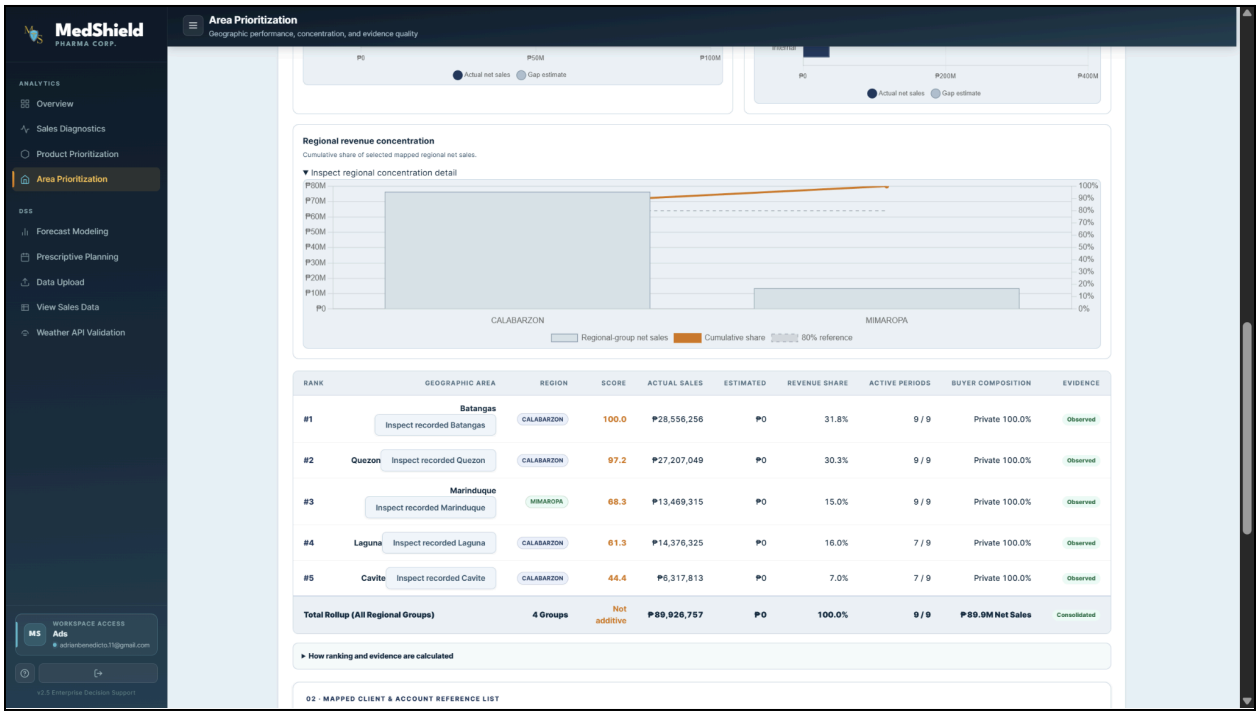
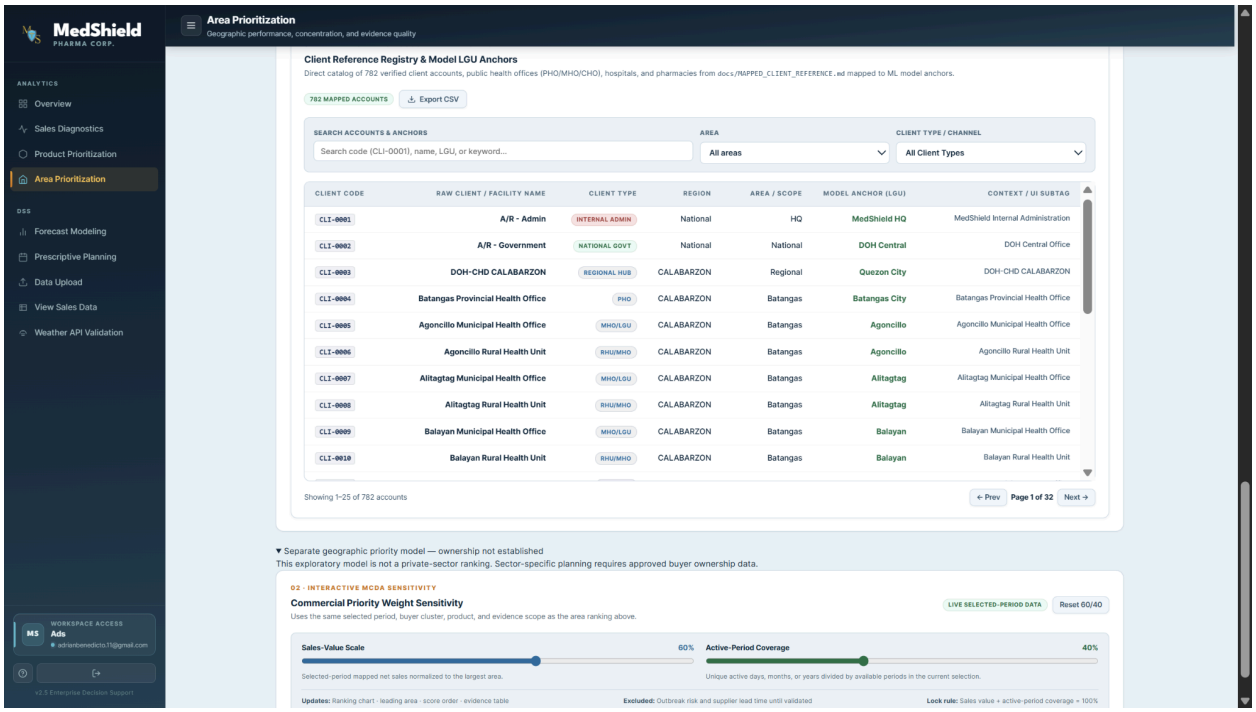


Figure 4.16 Area Ranking Details

The prioritization scores allowed users to examine the contribution of sales value and active-period coverage to each area's ranking. Meanwhile, the distinction between actual and estimated sales helped clarify the reliability of the information used in the analysis. Areas with higher sales contributions and more consistent historical activity could be considered for further inventory planning and distribution assessment. However, the rankings remained preliminary because geographic mappings were incomplete and some sales records could not be reliably attributed to specific territories. Consequently, the results supported the identification of potential priority areas for management review rather than definitive recommendations for inventory allocation or regional expansion.

4.3.4.3 Client Reference Registry and MCDA Sensitivity Scaling

Meanwhile, Figure 4.17 presents the Client Reference Registry. The registry contained 782 mapped client and account reference entries, with searchable fields for client codes, facility names, client types, geographic areas, and associated local government unit anchors.



The registry supported the organization of customer-related location information. However, the existence of mapped reference entries did not establish complete geographic attribution of all historical sales records. The relationship between transaction-level buyers and their corresponding geographic locations required further validation. The module also provided an interactive multicriteria decision analysis (MCDA) sensitivity interface. The displayed controls allowed the user to adjust the relative weight of sales value and active-period coverage. The default visible weighting assigned 60% to sales-value scale and 40% to active-period coverage, with the resulting scores and rankings intended to update according to the selected weights (Ng, 2007).

The interactive weighting function demonstrated how area prioritization could be examined under alternative decision criteria. Nevertheless, the displayed criteria did not include validated disease risk, supplier lead times, or complete operational inventory requirements. The resulting scores were therefore interpreted as exploratory geographic priorities rather than approved allocation decisions.

4.3.5 Forecast Modeling Module

The Forecast Modeling Module was developed to present historical sales observations,

forecasting benchmarks, model evaluation results, and exploratory external-variable analysis. The interface provided a visual comparison between observed monthly sales and candidate forecast outputs, supported by numerical accuracy indicators and detailed backtesting records (Box & Jenkins, 1970; Hyndman & Athanasopoulos, 2021).

4.3.5.1 Historical Backtesting and Model Comparison Charts

Figure 4.17 presents the Forecast Modeling Module, including a historical forecasting chart, evaluation metrics, and a comparison of baseline forecasting methods.

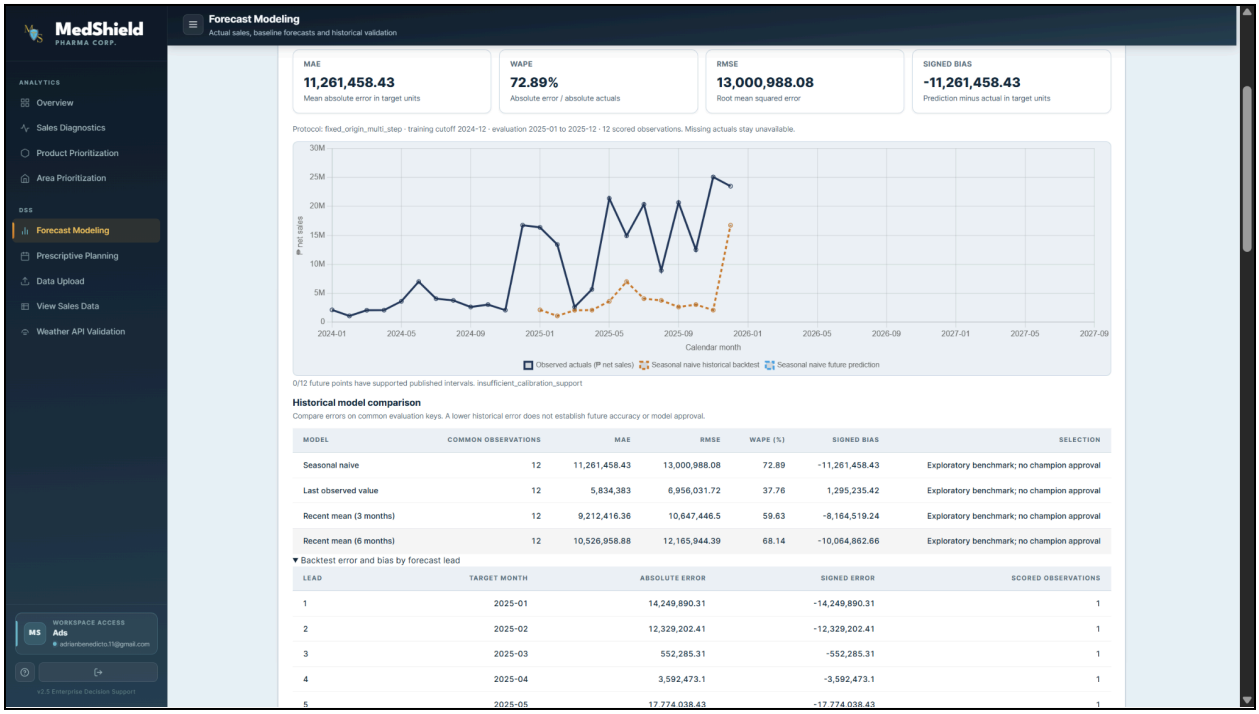


Figure 4.17. Forecast Modeling Module Showing Historical Backtesting and Model Comparison

The interface displayed mean absolute error (MAE), weighted absolute percentage error (WAPE), root mean squared error (RMSE), and signed bias for the selected evaluation. The displayed seasonal naive evaluation reported a WAPE of 72.89% for the 2025 audit period. However, the historical model comparison table showed lower error for the last-observed-value benchmark.

Table 4.11. Historical Forecasting Benchmark Comparison for the 2025 Audit Period

Forecasting Method	MAE (₱)	RMSE (₱)	WAPE (%)

Seasonal naive	11,261,458.43	13,000,988.08	72.89
Last observed value	5,834,383.00	6,956,031.72	37.76
Recent mean (3 months)	9,212,416.36	10,647,446.50	59.63
Recent mean (6 months)	10,526,958.88	12,165,944.39	68.14

The displayed benchmark comparison used 12 common observations for the 2025 audit period. Lower errors indicated better historical evaluation performance. The results did not establish operational approval or verified future forecasting accuracy. The last-observed-value method achieved the lowest WAPE among the four methods shown. The finding indicated that the comparatively simple benchmark performed better than the other displayed methods during the evaluated period. However, the remaining WAPE of 37.76% represented substantial forecast error and limited the suitability of the output for direct purchasing decisions. The Forecast Modeling Module also displayed forecast-lead information, monthly absolute errors, signed errors, and historical observation records. The detailed evaluation table provided greater transparency into the timing and magnitude of forecasting errors, particularly during months with unusually high sales.

The observed increase in sales during 2025 complicated the evaluation of forecasting methods trained on earlier historical patterns. Sudden operational shifts suggested historical relationships might not remain stable across distinct timeframes (Wolny & Kmiecik, 2024). Nevertheless, evaluation procedures did not confirm a structural break or isolate a single verified cause for the change.

The interface also identified the availability of prediction intervals. The captured evaluation indicated insufficient calibration support for publishing supported future prediction intervals. Consequently, the forecasting visualization remained an evaluation tool rather than evidence of an approved operational demand forecast. The module contributed to the predictive objective by demonstrating forecast evaluation, model comparison, and error reporting. The primary achievement was the establishment of an evidence-based process for assessing predictive reliability. Although the forecasting component could produce and evaluate candidate estimates, the available validation results did not justify presenting any method as a sufficiently reliable operational demand-planning model.

4.3.5.2 Disease and Weather Regression Evaluation

The Forecast Modeling Module contained an additional interface for examining the potential contribution of disease and meteorological observations to sales prediction. The component allowed users to select a buyer cluster, territory, sales measure, product scope, disease type, disease lag, weather source, and rainfall lag. Figure 4.18 presents the Disease and Weather Regression interface. The displayed selection included dengue as a candidate disease indicator, a one-month disease lag, and PAGASA meteorological observations as the weather source.

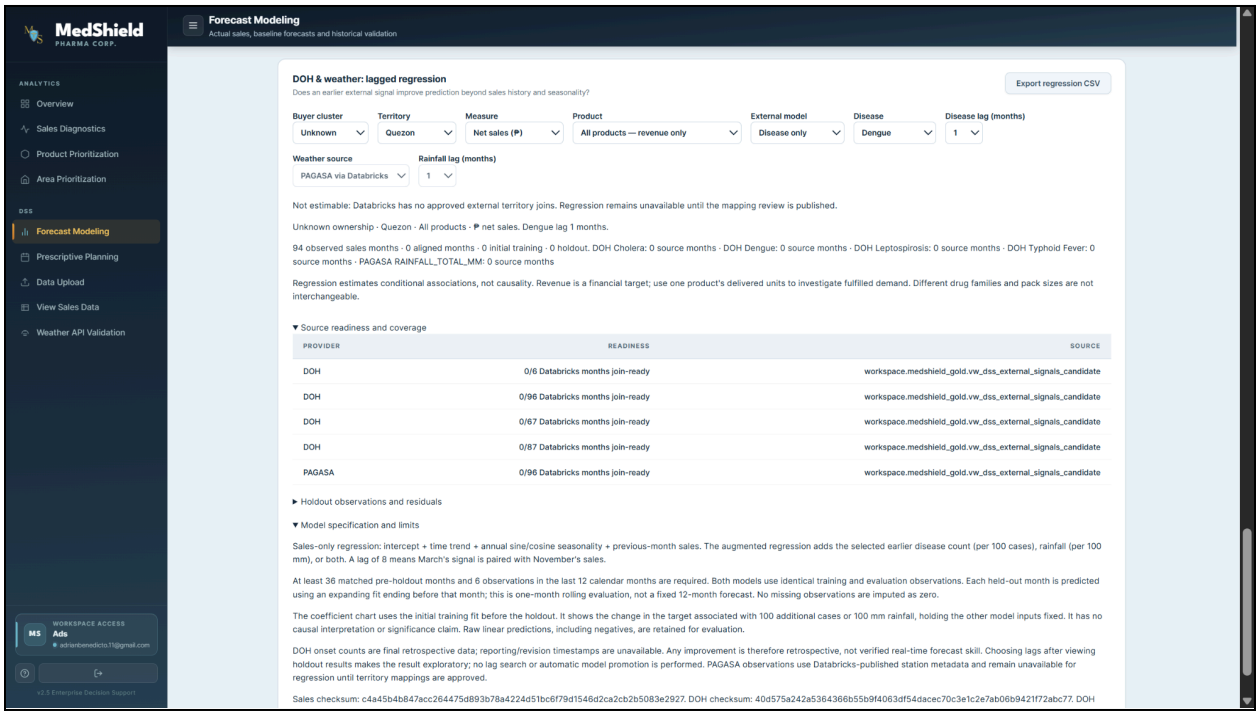


Figure 4.18. Disease and Weather Regression Interface and Data Readiness Status

The interface was designed to compare sales-only regression with models incorporating candidate external variables. The intended evaluation would determine whether earlier disease incidence or rainfall observations improved sales prediction beyond historical sales and seasonal effects. However, the displayed regression remained unavailable because approved territory-level external-data joins had not been established. The interface reported zero aligned months for the selected combination and indicated insufficient observations for model training or holdout evaluation.

The findings reinforced the integration limitations identified during data preparation. Although DOH epidemiological records and PAGASA meteorological observations were available as separate datasets, the existence of external observations did not establish a valid statistical relationship with MedShield's sales. The regression component also distinguished statistical association from causation. A regression coefficient, if successfully estimated, would describe a conditional relationship under the

specified model rather than establish that a disease outbreak or rainfall event directly caused a change in pharmaceutical demand.

In the captured implementation, no statistically supported disease–sales or rainfall–sales association could be reported for the selected territory and product scope. The external-variable component therefore remained exploratory and did not support attributing the unusual 2025 sales increase to disease outbreaks, typhoons, or other meteorological conditions. The regression interface contributed to the study by demonstrating the intended analytical workflow and identifying data-readiness requirements for future external-variable modeling. Approved geographic mappings, sufficient aligned observations, and comparative validation remain essential before disease and weather indicators can support operational forecasting.

4.3.6 Prescriptive Planning Module

The Prescriptive Planning Module was designed to support inventory-related decision-making through constrained allocation scenarios and user-supplied planning assumptions. The interface presented historically contributing products alongside input fields for projected requirements, available stock, safety reserves, pack quantities, procurement costs, and supplier limitations (Harris, 1913; Lepenioti et al., 2019; Puruhito & Falani, 2021).

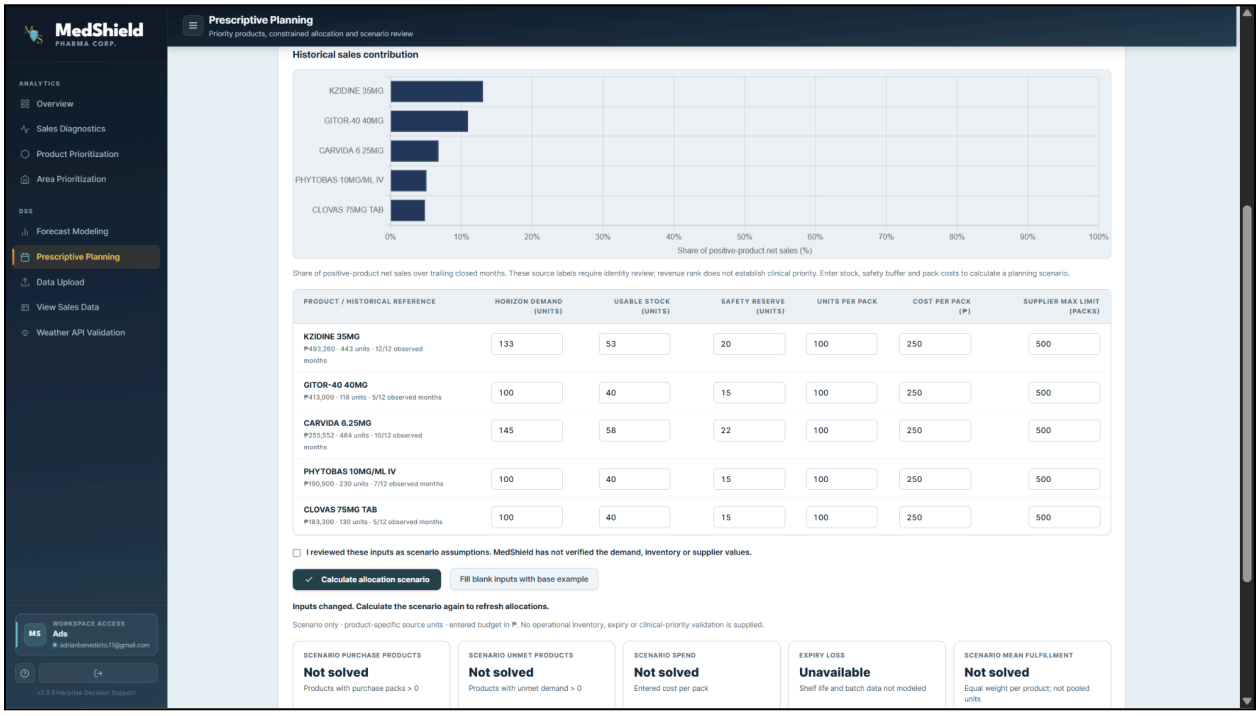


Figure 4.19. Prescriptive Planning Module and Scenario Input Interface

Figure 4.19 presents the Prescriptive Planning Module, including the historical product

contribution chart and the scenario input table. The product chart provided commercial context for candidate planning items, while the input table allowed users to enter operational assumptions for selected products. The displayed planning fields included horizon demand, usable stock, safety reserve, units per pack, cost per pack, and supplier maximum order limits. The combination of inputs established the basic parameters required to evaluate potential purchasing quantities and allocation constraints.

An acknowledgment control required users to indicate that the entered values represented reviewed scenario assumptions. The interface explicitly distinguished scenario parameters from verified operational information supplied by MedShield.

The module also provided a calculation control for evaluating an allocation scenario. Intended outputs included the number of products receiving purchase allocations, the number of products with unmet requirements, scenario spending, expiry-loss information, and mean fulfillment. In the captured interface, the allocation outputs were marked as **Not solved**, while expiry loss was marked **Unavailable**. The displayed state indicated that no completed allocation result could be reported from the screenshot. The expiry-loss indicator also identified the absence of modeled batch-level shelf-life and inventory information. The module demonstrated the structure of a prescriptive decision-support workflow by connecting selected products, user-defined constraints, and intended allocation outputs. However, actual procurement recommendations could not be established without verified inventory balances, supplier information, lead times, relevant costs, and batch-level expiry records.

The Prescriptive Planning Module contributed to the prescriptive objective at the scenario-design and calculation-interface level. The implementation demonstrated how management could examine alternative planning assumptions after the required inputs became available. Nevertheless, the captured interface did not establish an optimized purchasing plan, a confirmed allocation outcome, or a measured reduction in inventory losses.

4.3.7 Data Management Modules

Supporting data-management interfaces were incorporated into the DSS to improve transparency, source traceability, and verification of analytical records. The interfaces included Data Upload, View Sales Data, and Weather API Validation (Kimball & Ross, 2013; Oguntiloye, 2025).

4.3.7.1 Data Upload

The Data Upload screen identified Databricks Gold candidate views as the active analytical source and provided controls for uploading sales-related CSV and JSON data structures. The interface also displayed a Data Ingestion Audit Log and information concerning the active historical sales window.

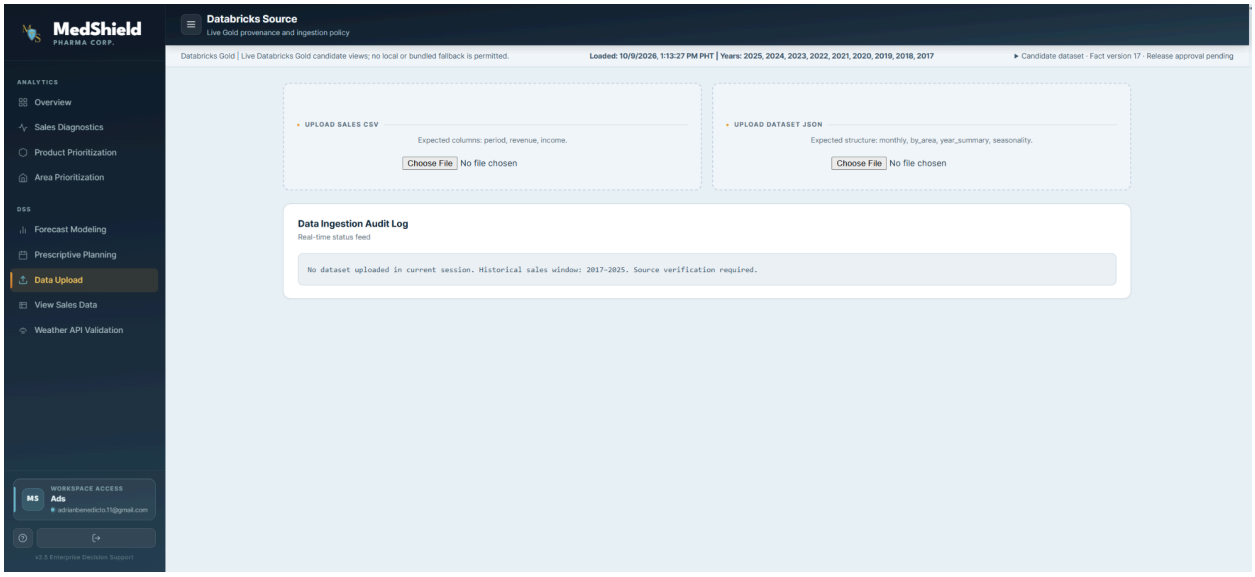


Figure 4.20. Databricks Source and Data Upload Interface

The source-status information identified the dataset as a candidate version awaiting release approval. The classification established an important distinction between records available for analytical review and records formally approved for operational publication.

4.3.7.2 View Sales Data

Figure 4.21 presents the View Sales Data interface, which displayed the accepted sales records through a structured analytical table. The interface contained transaction date, area, product description, source units sold, net sales revenue, acquisition cost, gross profit, and data-quality indicators.

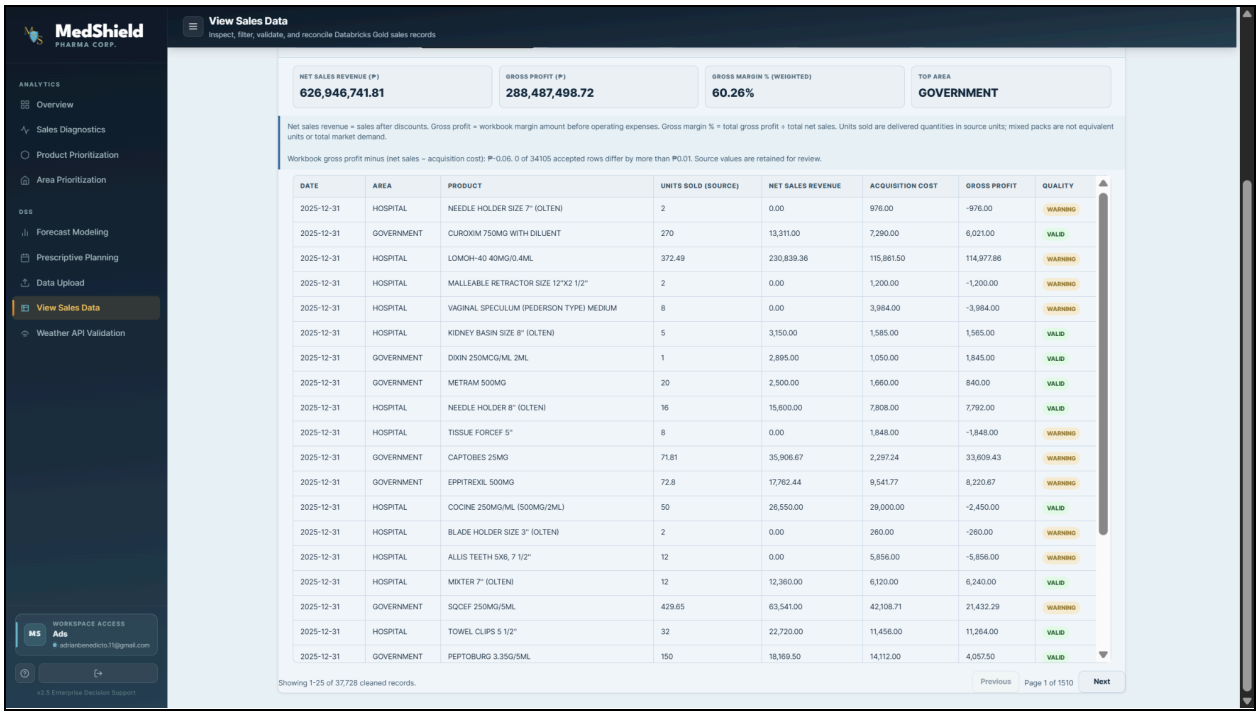


Figure 4.21. View Sales Data Interface Showing Transaction-Level Quality Indicators

The displayed table provided access to 37,728 cleaned sales records. Summary indicators reported ₱626,946,741.81 in net sales revenue, ₱288,487,498.72 in gross profit, and a weighted gross margin of 60.26%. Individual transactions were labeled with quality indicators, including valid and warning statuses. The warning labels supported the identification of records requiring review without automatically removing their values from the displayed analytical records. The interface also clarified the financial meaning of reported measures. Net sales represented revenue after discounts, while gross profit represented the workbook's margin amount before operating expenses. The displayed source quantities retained their original units and were not treated as standardized medicine demand quantities because products differed in unit definitions and packaging.

The View Sales Data interface contributed to the reliability of the DSS by allowing management and analysts to inspect the underlying transactions rather than relying solely on aggregate charts. The capability supported data reconciliation and the investigation of unusual revenue or margin values.

4.3.7.3 Weather API Validation

Figure 4.22 presents the Weather API Validation interface. The module displayed official PAGASA historical observations and separately identified NASA POWER proxy weather data. Users could select the source provider, territory, year, and available validation grain.

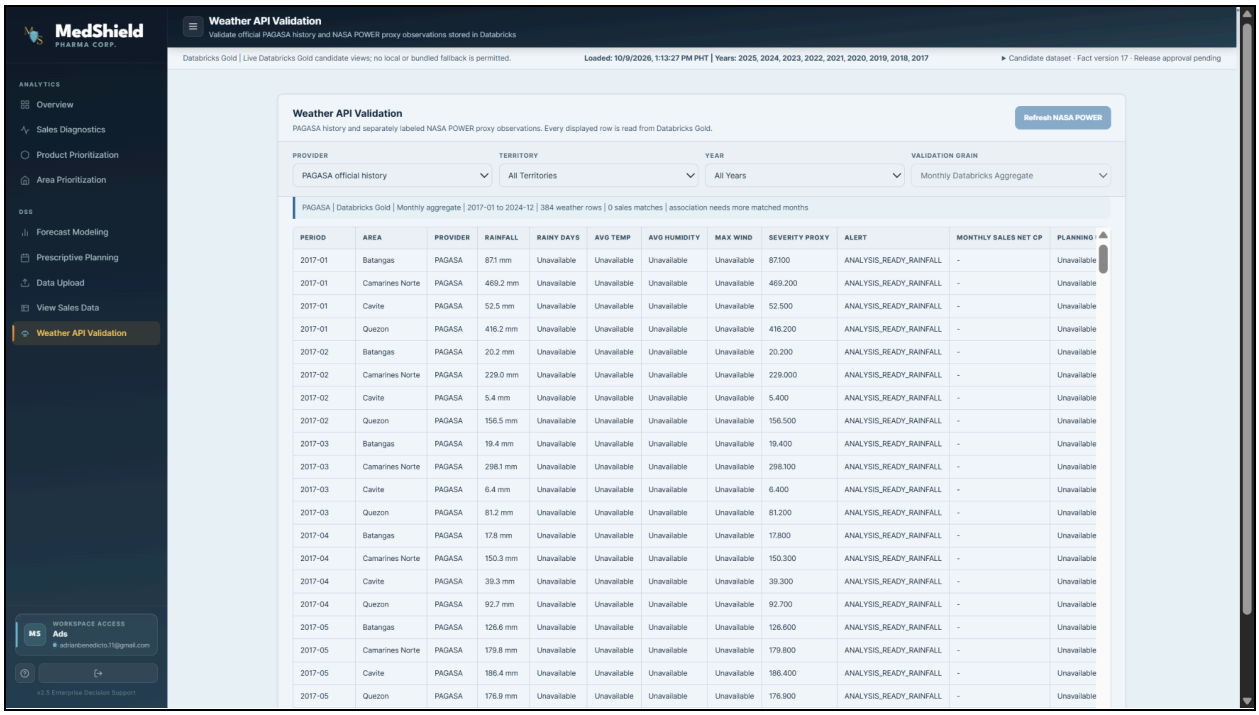


Figure 4.22. Weather API Validation Interface Showing PAGASA Records

The captured PAGASA view showed monthly meteorological observations for selected geographic areas, including rainfall values and available weather-related indicators. The interface also displayed information concerning geographic matching with monthly sales records. The selected validation view reported no matched sales records for the displayed weather observations. The result indicated that the existence of processed meteorological data alone was insufficient to establish a joined weather–sales analytical dataset.

The Weather API Validation interface therefore contributed to source verification and data-readiness assessment. By distinguishing official observations from proxy weather data and displaying integration limitations, the module reduced the risk of presenting incomplete geographic associations as established analytical findings. Together, the supporting data-management interfaces strengthened the traceability of the implemented DSS. Source identification, transaction-level examination, quality warnings, and external-data readiness indicators provided a basis for evaluating whether displayed analytical outputs were sufficiently supported by available records.

4.3.8 Dashboard Evaluation and Alignment with the Study Objectives

The evaluation of the developed DSS considered the implemented analytical functions, the evidence available through the dashboard, and the extent to which the modules addressed the study's five objective areas. The evaluation distinguished implemented functionality from analytical validation and confirmed operational outcomes.

Table 4.12. Dashboard Implementation and Alignment with the Study Objectives

Objective Area	Implemented DSS Component	Observed Result	Evaluation
Data preparation and integration	Data Upload, View Sales Data, Weather API Validation	Cleaned sales records, source traceability, quality indicators, and external-data readiness checks	Implemented for available records; some mappings remain incomplete
Descriptive analysis	Executive Overview, Sales Diagnostics, Product Prioritization, Area Prioritization	Historical sales trends, product contribution, candidate ABC, and geographic concentration	Supported by observed sales; classifications remain subject to limitations
Predictive analysis	Forecast Modeling and external-variable regression	Baseline comparison, backtesting, forecast errors, and external integration checks	Evaluated; no operational forecast approved
Prescriptive analysis	Prescriptive Planning	Scenario input interface and allocation-planning structure	Scenario-oriented; operational outcomes not established
DSS dashboard implementation and evaluation	Integrated dashboard and supporting interfaces	Centralized navigation, analytical visualizations, source-status information, and validation indicators	Implemented interface demonstrated; formal client acceptance evidence remains necessary

The descriptive components provided the clearest implemented analytical results. The Executive Overview and Sales Diagnostic Module displayed observed financial measures and identified historical changes in sales performance. The Product Prioritization Module supported preliminary sales-based ranking, while the Area Prioritization Module displayed geographic concentration and the limitations of available location mappings.

The predictive component demonstrated forecasting evaluation rather than operational forecasting success. The historical comparison identified the last-observed-value method as the best-performing displayed benchmark, although the remaining forecasting error was substantial. The disease and weather regression interface also identified insufficient approved integration data for the selected analysis. Consequently, predictive objectives were addressed through implementation and evaluation, but the evidence did not establish a production-ready model.

The prescriptive component demonstrated a planning interface designed for evaluating inventory allocation under user-defined constraints. However, the captured outputs did not contain solved procurement recommendations or verified expiry-loss calculations. Prescriptive functionality was therefore treated as scenario-oriented and dependent on additional operational inputs. The supporting data-management interfaces improved the transparency of the analytical workflow. Source-status information, transaction-quality indicators, and weather validation records allowed users to identify limitations affecting the interpretation of dashboard outputs.

The available screenshots demonstrated the presence and visible operation of major interface components. However, screenshots alone did not provide evidence of completed end-to-end functional testing, formal user acceptance testing, sustained production use, or actual improvements in inventory performance. Additional testing records and documented client feedback would be required to establish the corresponding evaluation outcomes.

4.3.10 Contribution of the DSS to Inventory Optimization and Loss Reduction

The central research concern of the study was how MedShield Pharma Corp. could optimize inventory and reduce losses through a decision support system. The developed dashboard addressed the concern by organizing historical evidence, identifying potential planning priorities, evaluating predictive reliability, and providing a structure for scenario-based inventory decisions.

First, the descriptive modules improved the visibility of historical sales performance. The annual and monthly visualizations allowed unusual changes in revenue to be identified, while product contribution analysis supported the preliminary identification of commercially significant products (Juran, 1954). The geographic module also provided information concerning sales concentration in mapped territories (Amuta et al., 2022). Second, the predictive module supported a more evidence-based approach to

forecasting. Rather than selecting an advanced method solely because of its complexity, the system displayed comparative forecasting errors and indicated the absence of an approved operational forecast (Hyndman & Athanasopoulos, 2021). The approach helped identify the level of uncertainty associated with possible demand-planning decisions. Third, the prescriptive module demonstrated the parameters required for constrained inventory allocation. Horizon demand, usable stock, safety reserves, pack quantities, procurement costs, and supplier limitations were represented as planning inputs (Harris, 1913; Shang et al., 2008). The module illustrated how inputs of such nature could be incorporated into future purchasing decisions once the values had been validated. Fourth, the supporting interfaces emphasized data quality and analytical accountability. Missing sales periods, source-product identity limitations, incomplete geographic mappings, and unavailable external-data joins were made visible rather than concealed through unsupported estimates.

The combined implementation established a foundation for improving inventory decisions (Attadjei et al., 2018; Woosley et al., 2009). Nevertheless, available evidence did not demonstrate an actual reduction in expired stock, overstocking, stockouts, procurement spending, or other operational inventory losses (Guadie et al., 2023). Direct measurement of operational outcomes requires pre-implementation and post-implementation inventory records, verified stock and expiry data, and documented use of approved recommendations. Overall, the DSS demonstrated how descriptive analysis, forecasting evaluation, and scenario-based planning could be organized into an integrated decision-support environment. The implemented system advanced MedShield's ability to examine historical commercial activity and identify the evidence needed for inventory optimization. Its contribution to actual loss reduction remained a potential operational benefit rather than a verified outcome.

4.4 Summary of Findings

Chapter 5
Conclusion and Recommendation

The chapter presents the conclusions drawn from the development, implementation, and empirical evaluation of the MedShield Pharma Corp. Decision Support System (DSS), followed by actionable recommendations for its future application. By synthesizing the findings of the study, it evaluated the extent to which the system achieved its primary objectives of improving inventory stock management and optimizing regional distribution planning. Furthermore, the insights derived from the system's performance were translated into strategic, tactical, and competitive recommendations that can support MedShield Pharma Corp. in addressing operational challenges, including seasonal weather variations and disease outbreaks. These recommendations were aimed to strengthen supply chain resilience and promote more effective data-driven decision-making.

5.1 Conclusions

Summarize the main points of your study. State clearly whether your project achieved its objectives and affirm the significance of your results in relation to the original problem statement.

5.2 Recommendations

Propose actionable next steps. These should be based on your findings and demonstrate how the project can be taken further.

5.2.1 For Strategic Planning

Suggest how long-term organizational or project goals can be aligned with or improved by your system.

5.2.2 For Competitive Advantage

Discuss how your solution offers a unique benefit or edge over existing methods or competitors.

5.2.3 For Tactical Decisions

Recommend specific, immediate operational changes or adjustments that users can implement using your system on a daily or short-term basis.

References

Abrigo, M. R. M., Opiniano, G. A., Tam, Z. C., Yee, S. A., & Estopace, K. M. M. (2021). Public sector procurement of medicines in the Philippines. In EconStor. Philippine Institute for Development Studies. https://hdl.handle.net/10419/256879

Amuta, M. A., Muonde, M., Mustapha, A. Y., & Mbata, A. O. (2022). A decision-support framework for allocating pharmaceutical sales territories based on epidemiological and market data. Gyanshauryam, International Scientific Refereed Research Journal, 5(5). https://gisrrj.com/paper/GISRRJ2256126.pdf

Anand, L. G. (2025). Using Machine Learning in SAP IBP for Supply Chain Optimization. In Social Science Research Network. Elsevier BV. https://doi.org/10.2139/ssrn.5240827

Attadjei, D. D., Madhwal, Y., & Panfilov, P. B. (2018). A decision phases of a supply chain management: A proposed decision support system to boost organizational decision making. International Journal of Engineering & Technology, 7(2.28), 157. https://doi.org/10.14419/ijet.v7i2.28.12901

Azevedo, A., & Santos, M. F. (2008). KDD, SEMMA, and CRISP-DM: A parallel overview. IADIS European Conference Data Mining. https://www.iadisportal.org/digital-library/kdd-semma-and-crisp-dm-a-parallel-overview

Bacani, N. D., Balitustus, D. D. C., Porcalla, C. S., Sison, A. A. R. C., Guialil, J. S., & Centeno, C. J. (2024). BOTika: A pharmaceutical management system using seasonal autoregressive integrated moving average (SARIMA) and clustering algorithm. International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems, 1–9. https://doi.org/10.1109/icses63760.2024.10910563

Ball, D., & Salenga, R. (2017). Pharmaceutical policy in the Philippines. In Z. U. D. Babar (Ed.), Pharmaceutical policy in countries with developing healthcare systems. Adis. https://doi.org/10.1007/978-3-319-51673-8_4

Belghith, M., Ben Ammar, H., Elloumi, A., & Hachicha, W. (2023). A new rolling forecasting framework using Microsoft Power BI for data visualization: A case study in a pharmaceutical industry. Annales Pharmaceutiques Françaises.

Bejarano, A., Nadungodage, C. H., Wang, F., Catlin, A. C., & Hoag, S. W. (2019). Decision support for excipient risk assessment in pharmaceutical manufacturing. AAPS PharmSciTech, 20(6), 223. https://doi.org/10.1208/s12249-019-1440-x

Bello, S. S., Benjamin, U. I., & Ejiako, M. O. (2026). Development of a decision support system for inventory management in a pharmaceutical retail outlet. World Scientific News, 211(2026). https://worldscientificnews.com/wp-content/uploads/2026/01/WSN-211-2026-108-116.pdf

Board of Investments. (2025, September 22). BOI highlights implementation of PH Pharma Industry Roadmap at PCPI's 2nd GMM 2025. https://boi.gov.ph/boi-highlights-implementation-of-ph-pharma-industry-roadmap-at-pcpis-2nd-gmm-2025/

Bolikulov, F., Nasimov, R., Rashidov, A., Akhmedov, F., & Cho, Y.-I. (2024). Effective methods of categorical data encoding for artificial intelligence algorithms. Mathematics, 12(16), 2553. https://doi.org/10.3390/math12162553

Box, G. E. P., & Jenkins, G. M. (1970). Time series analysis: Forecasting and control. Holden-Day. https://www.wiley.com/en-us/Time+Series+Analysis%3A+Forecasting+and+Control%2C+5th+Edition-p-9781118675021

Burinskiene, A. (2022). Forecasting models for pharmaceutical retail. Frontiers in Medicine, 9, 582186. https://doi.org/10.3389/fmed.2022.582186

Candeias, L. D. A. (2024). Pharmaceutical Sales Representatives' Performance and Impact on Revenue - Using a Data Mining Approach. Handle.net. http://hdl.handle.net/10362/164187

Chango, W., Salguero, A., Landivar, T., Vásconez, R., Silva, G., Peñafiel-Arcos, P., Núñez, L., & Velasteguí-Izurieta, H. (2025). SARIMA vs. Prophet: Comparative efficacy in forecasting traffic accidents across Ecuadorian provinces. Computation, 14(1), 5. https://doi.org/10.3390/computation14010005

Chapman, P., Clinton, J., Kerber, R., Khabaza, T., Reinartz, T., Shearer, C., & Wirth, R. (2000). CRISP-DM 1.0: step-by-step data mining guide. SPSS. https://www.kde.cs.uni-kassel.de/wp-content/uploads/lehre/ws2012-13/kdd/files/CRISPWP-0800.pdf

Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. ArXiv (Cornell University). https://doi.org/10.48550/arxiv.1603.02754

Cleveland, R. B., Cleveland, W. S., McRae, J. E., & Terpenning, I. (1990). STL: A

seasonal-trend decomposition procedure based on Loess. Journal of Official Statistics, 6(1), 3–73. https://www.stats.govt.nz/assets/Uploads/Methods/Documents/3-1-cleveland-et-al-STL.pdf

Davis, J., & Goadrich, M. (2006). The relationship between precision-recall and ROC curves. Proceedings of the 23rd International Conference on Machine Learning, 233–240. https://doi.org/10.1145/1143844.1143874

Dela Cruz, J., Gengania, R. K., Nacar, C. R., & Rusiana, F. (2025). Data-driven approach in forecasting of medical supply shortages and overstock: A case study of SLU-Sacred Heart Medical Center (SLU-SHMC). Journal of Global Business, 14(2). https://atiftap.org/s/JGB-19216.pdf

Dong, G., & Liu, H. (Eds.). (2018). Feature engineering for machine learning and data analytics. CRC Press. https://doi.org/10.1201/9781315181080

Dy-Liacco, C. M. M., Magsajo, T. D. C., Say, K. S., Go, C. K. C., Uy, J., & Antonio, V. A. A. (2024). Modeling pooled procurement of Philippine Department of Health drugs and medicines using OLS regression. AIP Conference Proceedings, 3132(30002). https://doi.org/10.1063/5.0213405

Fahad, A., & Kollwitz, E. (2025). The digital transformation of sales: Examining the role of technology adoption in sales enablement. https://doi.org/10.13140/rg.2.2.11555.36640

Fattah, J., Ezzine, L., Aman, Z., El Moussami, H., & Lachhab, A. (2018). Forecasting of demand using ARIMA model. International Journal of Engineering Business Management, 10. https://doi.org/10.1177/1847979018808673

Federal Democratic Republic of Ethiopia, Ministry of Health. (2015). Health sector transformation plan (HSTP) 2015/16–2019/20. Federal Democratic Republic of Ethiopia, Ministry of Health. https://www.issup.net/knowledge-share/resources/2021-05/health-sector-transformation-plan-201516-201920

Fernando, J. G., & Baldelovar, M. (2022). Decision support system: Overview, different types, and elements. Technoarete Transactions on Intelligent Data Mining and Knowledge Discovery, 2(2). https://doi.org/10.36647/ttidmkd/02.02.a003

Firas, O. (2023). A combination of SEMMA & CRISP-DM models for effectively handling big

data using formal concept analysis based knowledge discovery: A data mining approach. World Journal of Advanced Engineering Technology and Sciences, 8(1), 009-014. https://doi.org/10.30574/wjaets.2023.8.1.0147

Flores, B. E., & Whybark, D. C. (1986). Multiple criteria ABC analysis. International Journal of Operations & Production Management, 6(3), 38–46. https://doi.org/10.1108/eb054719

Guadie, M., Asemahagn, M. A., Tefera, A., Melkam, W., Habteweld, H. A., & Derebe, D. (2023). Medicines wastage and its contributing factors in public health facilities of South Gondar Zone, Amhara Regional State, Ethiopia. Integrated Pharmacy Research and Practice, Volume 12(12), 157–170. https://doi.org/10.2147/iprp.s415375

Giebel, G. D., Raszke, P., Nowak, H., Palmowski, L., Adamzik, M., Heinz, P., Tokic, M., Timmesfeld, N., Brunkhorst, F., Wasem, J., & Blase, N. (2024). Problems and barriers related to the use of AI-based clinical decision support systems: Interview study. Journal of Medical Internet Research, 27, e63377. https://doi.org/10.2196/63377

Halim, I., Ang, P., & Adhitya, A. (2019). A decision support framework and system for design of sustainable pharmaceutical supply chain network. Clean Technologies and Environmental Policy, 21(2), 431–446. https://doi.org/10.1007/s10098-018-1646-8

Harris, F. W. (1913). How many parts to make at once. Operations Research, 38(6), 947–950. https://doi.org/10.1287/opre.38.6.947

İmece, S., & Beyca, Ö. F. (2022). Demand forecasting with integration of time series and regression models in pharmaceutical industry. International Journal of Advances in Engineering and Pure Sciences, 34(3), 415–425. https://doi.org/10.7240/jeps.1127844

Juran, J. M. (1954). Universals in management planning and controlling. The Management Review, 43(11), 748–761.

Juran, J. M. (1975). The non-Pareto principle; mea culpa. Quality Progress, 8(5), 8–9. https://www.juran.com/wp-content/uploads/2021/03/The-Non-Pareto-Principle-1974.pdf

Holt, C. C. (2004). Forecasting seasonals and trends by exponentially weighted moving averages. International Journal of Forecasting, 20(1), 5–10. https://doi.org/10.1016/j.ijforecast.2003.09.015 (Original work published 1957)

Hyndman, R. J., & Athanasopoulos, G. (2021). Forecasting: Principles and Practice (3rd ed.). OTexts. https://otexts.com/fpp3/

Kimball, R., & Ross, M. (2013). The data warehouse toolkit: The definitive guide to dimensional

modeling (3rd ed.). John Wiley & Sons, Inc. https://www.wiley.com/en-us/The+Data+Warehouse+Toolkit%3A+The+Definitive+Guide+to+Dimensional+Modeling%2C+3rd+Edition-p-9781118530801

Kwarteng, S., & Andreevich, P. A. (2024). Comparative analysis of ARIMA, SARIMA and Prophet model in forecasting. Research & Development, 5(4), 110–120. https://doi.org/10.11648/j.rd.20240504.13

Lepenioti, K., Bousdekis, A., Apostolou, D., & Mentzas, G. (2019). Prescriptive analytics: Literature review and research challenges. International Journal of Information Management, 50, 57–70. https://doi.org/10.1016/j.ijinfomgt.2019.04.00

Martinez, M. E. (2018). The calendar of epidemics: Seasonal cycles of infectious diseases. PLOS Pathogens, 14(11), e1007327. https://doi.org/10.1371/journal.ppat.1007327

Maulana, K. R., Widiyono, W., & Darmawan, A. S. (2025). Predicting tablet drug expenditures using Python-based Facebook Prophet in pharmaceutical installations. Jurnal Teknologi Dan Open Source, 8(2), 709–721. https://doi.org/10.36378/jtos.v8i2.4907

Mohamed, G. (2025). Comparative analysis of AI-driven decision support system and traditional spreadsheets: Evaluating accuracy and consistency in business intelligence. Journal of Science and Technology, 30(4), 80–89. https://doi.org/10.20428/jst.v30i4.2765

Ng, W. L. (2007). A simple classifier for multiple criteria ABC analysis. European Journal of Operational Research, 177(1), 344–353. https://doi.org/10.1016/j.ejor.2005.11.018

Nguyen, A., Lamouri, S., Pellerin, R., Tamayo, S., & Lekens, B. (2021). Data analytics in pharmaceutical supply chains: State of the art, opportunities, and challenges. International Journal of Production Research, 60(22), 6888–6907. https://doi.org/10.1080/00207543.2021.1950937

Oguntiloye, D. (2025). Data Analytics and Visualization Dashboard for Supply Chain and Procurement as a Performance Assessment Tool. International Journal of Innovative Science and Research Technology, 10(8), 1539–1555. https://doi.org/10.38124/ijisrt/25aug975

Palacios, H. J. G., Toledo, R. A. J., Pantoja, G. A. H., & Navarro, Á. A. M. (2017). A comparative between CRISP-DM and SEMMA through the construction of a MODIS repository for studies of land use and cover change. Advances in Science, Technology and Engineering Systems Journal, 2(3), 598–604. https://doi.org/10.25046/aj020376

Patoulia, A. A., Kiourtis, A., Mavrogiorgou, A., & Kyriazis, D. (2022). A comparative study of collaborative filtering in product recommendation. Emerging Science Journal, 7(1), 1–15. https://doi.org/10.28991/esj-2023-07-01-01

Primo, R. C., & Mariano, V. Y. (2025). AFProTrack: A longitudinal training trajectory management system for army personnel using multi-platform technologies. IEEE Symposium on Wireless Technology & Applications, 1–6. https://doi.org/10.1109/iswta68114.2025.11329565

Puruhito, M. A., & Falani, A. Z. (2021). Decision support system for developing application for pharmaceutical supplies using the MMSL and Pareto Law methods. International Journal of Electrical Engineering and Information Technology, 4(1), 12–22. https://doi.org/10.29138/ijeeit.v4i1.1329

Ramanathan, R. (2006). ABC inventory classification with multiple-criteria using weighted linear optimization. Computers & Operations Research, 33(3), 695–700. https://doi.org/10.1016/j.cor.2004.07.014

Rekabi, S., Sazvar, Z., & Shakibaei, H. (2025). Designing a sustainable-resilient pharmaceutical supply chain network using a machine learning-based approach. OPSEARCH. https://doi.org/10.1007/s12597-025-00954-6

Sari, I. U., & Yilmaz, Y. U. (2025). A decision support system for developing marketing strategy in pharmaceutical industry. International Symposium on Innovative Approaches in Smart Technologies, 1–6. https://doi.org/10.1109/isas66241.2025.11101819

SAS Institute Inc. (2025). SAS Enterprise Miner 15.4: Reference help. sas.com

Schröer, C., Kruse, F., & Gómez, J. M. (2021). A systematic literature review on applying CRISP-DM process model. Procedia Computer Science, 181(1), 526–534. https://doi.org/10.1016/j.procs.2021.01.199

Shimaoka, A. M., Ferreira, R. C., & Goldman, A. (2023). The evolution of CRISP-DM for Data Science: Methods, processes and frameworks. SBC Reviews on Computer Science, 1(1). https://doi.org/10.13140/RG.2.2.22493.42721

Shang, J., Tadikamalla, P. R., Kirsch, L. J., & Brown, L. (2008). A decision support system for managing inventory at GlaxoSmithKline. European Journal of Operational Research, 190(1), 222–235. https://doi.org/10.1016/j.ejor.2007.06.015

Silveira, R. F., Lima, A. L., Gross, I. P., Gelfuso, G. M., Gratieri, T., & Cunha-Filho, M. (2024).

The role of artificial intelligence and data science in nanoparticles development: a review. Nanomedicine, 19(14), 1271–1283. https://doi.org/10.1080/17435889.2024.2359355

Sokolova, M., & Lapalme, G. (2009). A Systematic Analysis of Performance Measures for Classification Tasks. Information Processing & Management, 45(4), 427–437. https://doi.org/10.1016/j.ipm.2009.03.002

Sun, Z. (2024). Data Analytics for Business Intelligence. CRC Press. https://lccn.loc.gov/202402956

Taylor, S. J., & Letham, B. (2018). Forecasting at Scale. The American Statistician, 72(1), 37–45.

https://doi.org/10.1080/00031305.2017.1380080

Tiongson, J. H. V., & Kummer, M. G. C. (2020). Rural health unit decision support system with mapping. International Journal of Recent Technology and Engineering, 9(3), 611–616. https://doi.org/10.35940/ijrte.c4681.099320

Umadevi, G., & Umamaheswari, S. (2023). Advancing healthcare service efficacy by optimizing pharmaceutical inventory management: Leveraging ABC, VED analysis for trend demand. International Journal of Statistics in Medical Research, 12, 283–293. https://doi.org/10.6000/1929-6029.2023.12.33

Williams, S. J. (2024). Supply Chain Management. Cambridge University Press. http://www.cambridge.org/9781009325288

Winters, P. R. (1960). Forecasting sales by exponentially weighted moving averages. Management Science, 6(3), 324–342. https://doi.org/10.1287/mnsc.6.3.324

Wolny, M., & Kmiecik, M. (2024). Unveiling Patterns in Forecasting Errors: A Case Study of 3PL Logistics in Pharmaceutical and Appliance Sectors. Sustainability, 17(1), 214–214. https://doi.org/10.3390/su17010214

Woosley, J. M., Wiley-Patton, S., Kelle, P., & Schneider, H. (2009). Decision support in healthcare supply chain management and pharmaceutical inventory control. AMCIS 2009 Proceedings, 522. https://aisel.aisnet.org/amcis2009/522

World Health Organization. (2024). Digital transformation handbook for health supply chain architecture. https://www.who.int/publications/i/item/9789240101197